

Prostate segmentation from T2-weighted MRI-images

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Introduction

Prostate cancer is one of the most common cancers among men worldwide, and accurate segmentation of the prostate gland from T2-weighted MRI images is crucial for diagnosis and treatment planning. In recent years, deep learning-based segmentation approaches have shown promising results in automating this task, offering potential benefits in terms of efficiency and accuracy.

In this project, we aim to compare the performance of three UNet-based segmentation models using different backbone architectures (VGG19, ResNet50, EfficientNetb0) on datasets obtained from two different medical imaging centers, referred to as Site A and Site B. The primary objectives of our investigation are to assess:

1. How each model performs on the datasets from Site A and Site B individually.
2. The ability of the best-performing model trained on one site to generalize to the test set from the other site (cross-site generalization).

By conducting this comparative analysis, we seek to gain insights into the effectiveness of different backbone architectures for prostate MRI segmentation and explore the challenges of deploying models trained on one dataset to another dataset from a different imaging center. Such insights can inform the development of more robust and reliable segmentation algorithms for clinical applications in prostate cancer diagnosis and treatment.

Dataset Description

The dataset used in this project was obtained from the Multi-site Dataset for Prostate MRI Segmentation, accessible from the link (<https://liuquande.github.io/SAML/>). This organized dataset comprises T2-weighted Prostate MRI data collected from six diverse data sources, representing different institutions and imaging protocols. To ensure consistency, preprocessing steps were uniformly applied to the data from all six sites, including converting the data to the '.nii' format and resizing all samples to a standardized size of 384x384 in the axial plane.

Site A and Site B

For our project, we focused on data from two sites, denoted as Site A and Site B. Each site contained data from 30 patients and exhibited distinct characteristics:

	Site A (RUNMC)	Site B (BMC)
Institution	Radboud University, Nijmegen Medical Centre	Boston University Medical Centre
Case Number	30	30
Field Strength (T)	3	1.5
Resolution (in/through plane) (mm)	0.6-0.625/3.6-4	0.4/3
Endorectal Coil	Surface	Endorectal
Manufacturer	Siemens	Philips

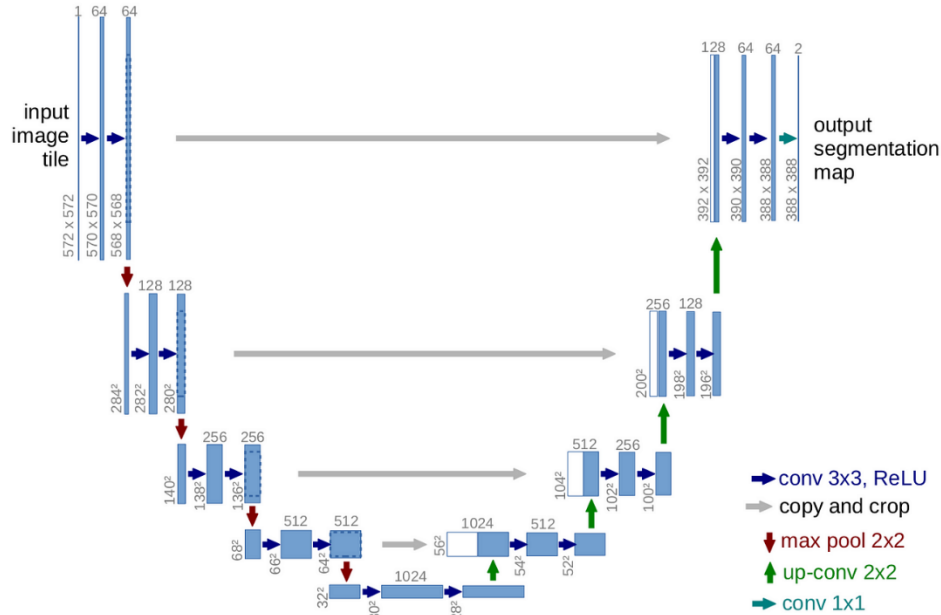
Key Parameters

- Field Strength (T):** Indicates the strength of the magnetic field used in the MRI scanner. Higher field strength, such as 3 T, can enhance the signal-to-noise ratio and image quality.
- Resolution (in/through plane) (mm):** Represents the level of detail in the MRI images. Smaller values in millimeters denote higher resolution, enabling clearer and more detailed images.
- Endorectal Coil:** A specialized coil inserted into the rectum during MRI examinations to enhance prostate imaging. Its use may vary depending on scanner type and preferences.

Methodology

In this section, we present an overview of the methodology employed in our study, focusing on the UNet architecture and the three different backbones used: VGG19, ResNet50, and EfficientNetB0.

UNet Architecture:



UNet is a popular convolutional neural network (CNN) architecture commonly used for image segmentation tasks. It consists of a contracting path to capture context and a symmetric expanding path that enables precise localization. The architecture comprises two main parts: the encoder and the decoder.

Encoder: The encoder downsamples the input image through a series of convolutional and pooling layers, gradually extracting hierarchical features.

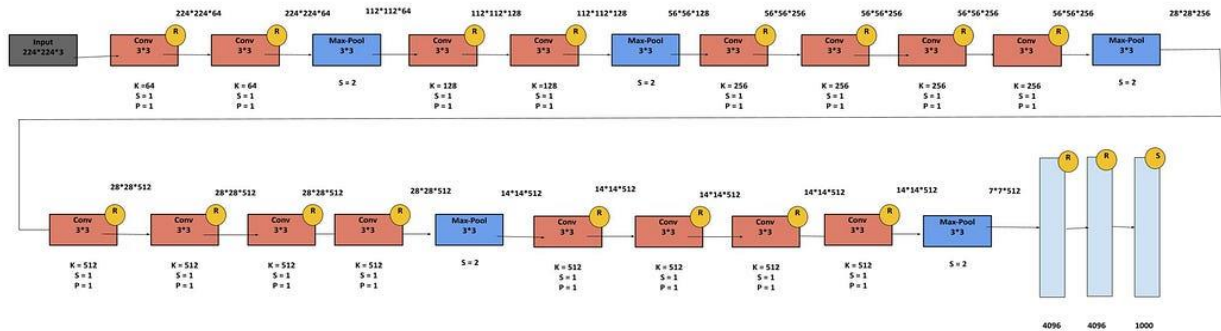
Decoder: The decoder upsamples the feature maps to the original input size, generating a segmentation mask with the help of skip connections that concatenate feature maps from the encoder.

UNet's unique design allows it to effectively capture both local and global information, making it well-suited for medical image segmentation tasks where precise delineation of structures is essential.

Backbone Architectures:

In this project, we assess the effectiveness of three different backbone architectures incorporated into the UNet framework. These architectures are utilized specifically for feature extraction within the encoder section of the UNet model, aiming to enhance prostate MRI segmentation.

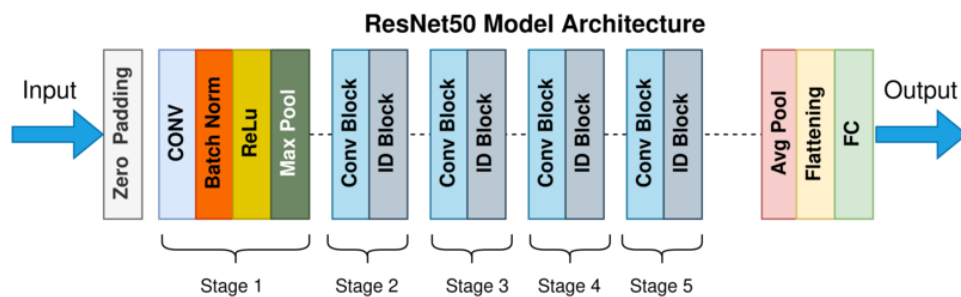
VGG19:



VGG19 is a convolutional neural network (CNN) architecture comprising 19 layers, including 16 convolutional layers, 3 fully connected layers, 5 max-pooling layers, and 1 softmax layer. It is an extension of the VGG model developed by the Visual Geometry Group at Oxford University. VGG19 achieved remarkable success in image classification tasks, particularly in the ImageNet Large-Scale Visual Recognition Challenge (ILSVRC) in 2014.

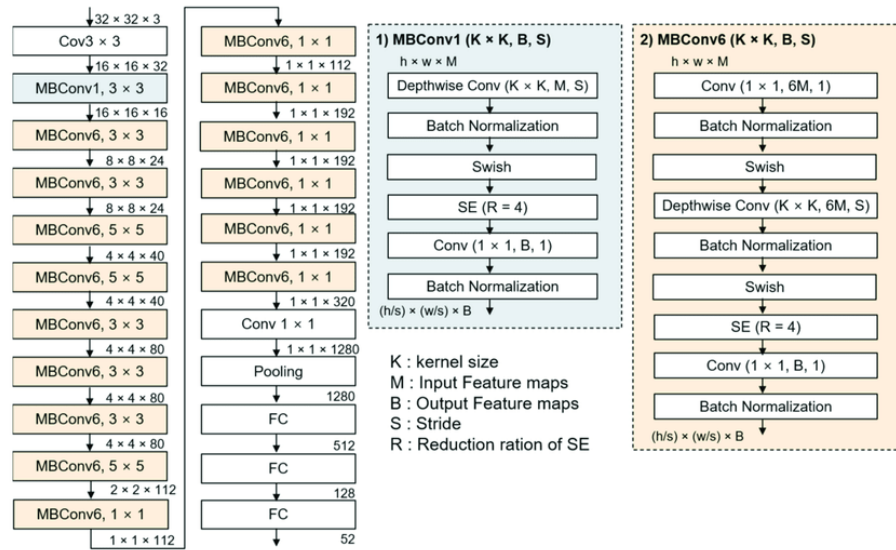
The architecture involves processing input images of size (224, 224, 3) through convolutional layers with 3x3 kernels and stride of 1 pixel, followed by max-pooling with 2x2 windows and stride of 2. Rectified linear units (ReLU) are used for non-linearity, enhancing classification accuracy and computational efficiency. The network concludes with three fully connected layers, the last of which employs softmax activation for 1000-way classification in ILSVRC. VGG19's effectiveness lies in its deep convolutional layers and the integration of ReLU activation functions, contributing to improved feature extraction and classification accuracy.

ResNet50:



ResNet50, short for "residual network, is a deep convolutional neural network architecture developed by Microsoft Research in 2015, featuring 50 layers. It incorporates residual connections to enable the learning of deeper architectures without the vanishing gradient problem. The architecture includes convolutional layers for feature extraction, identity blocks for learning residual functions, and convolutional blocks for transformation. Max pooling layers reduce spatial dimensions, and fully connected layers produce final classifications. ResNet50 is renowned for its effectiveness in image classification tasks due to its ability to learn intricate features from input images.

EfficientNetB0



EfficientNet is a convolutional neural network architecture that introduces a novel scaling method to uniformly adjust the network's depth, width, and resolution using a compound coefficient. Unlike conventional approaches that arbitrarily scale these factors, EfficientNet uses a systematic method to scale network dimensions based on a compound coefficient, enabling efficient use of computational resources while maintaining performance.

Model Training Configuration

Preprocessing:

Before feeding the images into the model, preprocessing steps were applied to standardize and prepare the data.

1. **Class Combination:** The two prostate zones, Central Gland (CG) and Peripheral Zone (PZ), were merged into a single class for binary segmentation.
2. **Slice Removal:** Slices with segmentation information covering less than 1% of the image were removed to focus on informative data.
3. **Channel Duplication:** Single-channel images were duplicated to three channels to match the requirements of the UNet model architecture.
4. **Resizing:** Images and masks were resized from (384,384) resulting in axial plane dimensions of (224, 224).
5. **Dataset Creation:** The final dataset was generated by iterating through all slices of the images and masks, ensuring compatibility with 2D UNet model architecture for segmentation tasks.

Common Parameters:

Loss Functions:

The loss function measures the inconsistency between the predicted output of a model and the true target values in a training dataset. It quantifies how well the model is performing relative to the ground

truth during training. The choice of loss function depends on the specific task and the type of model being trained. For the segmentation task in this project, we used two loss functions to calculate the total loss, the Dice Loss, and Binary Focal Loss.

Dice Loss:

Dice loss, also called the Sørensen-Dice coefficient, assesses the agreement between predicted and ground truth segmentation masks by measuring their overlap. It's computed as twice the intersection of the masks divided by the sum of their areas. Typically employed as a loss function during model training, Dice loss ranges from 0 to 1, with 0 indicating no overlap and 1 indicating perfect agreement. Notably, it equally penalizes false positives and false negatives, making it effective for optimizing segmentation models.

$$Dice = \frac{2 \times TP}{(TP + FP) + (TP + FN)}$$

Binary Focal Loss:

Binary Focal Loss addresses class imbalance by down-weighting well-classified examples and amplifying loss for hard, misclassified examples using a "focusing parameter." It's particularly useful in binary classification tasks with imbalanced datasets, where conventional loss functions may prioritize majority class examples over minority ones, leading to poor performance.

$$FL(p_t) = -\alpha_t(1 - p_t)^\gamma \log(p_t)$$

Where:

- p_t is the predicted probability of the true class,
- α_t is the class-balancing factor (typically used to counter class imbalance),
- γ is the focusing parameter that controls the rate at which easy examples are down-weighted compared to hard examples.

These two loss functions are combined using the + operator and scaled by a factor of 1 for the total loss.

Optimization:

The optimizer determines the update rules for the weights of the neural network during training.

In this project, the Adam optimizer is chosen with a learning rate of 0.0001. Adam is a popular choice for optimization due to its adaptive learning rate and momentum-based updates, which help in faster convergence and better performance.

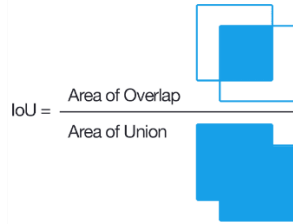
Metrics:

Metrics are used to evaluate the performance of the model during training and validation.

Two metrics are used in this configuration: Intersection over Union (IoU) Score and F1 Score.

IoU Score:

IoU Score, also known as Jaccard Index, measures the overlap between the predicted and ground truth segmentation masks, providing an indication of segmentation accuracy. The IOU score is calculated as the ratio of the intersection of the predicted and ground truth masks to the union of these masks. IOU score ranges from 0 to 1, where a score of 1 indicates perfect overlap between the predicted and ground truth masks, while a score of 0 indicates no overlap at all.



$$IoU = \frac{TP}{(TP + FP + FN)}$$

Where:

- TP - (True Positive) is the number of pixels that are correctly classified as positive (both in the predicted and ground truth masks).
- FP - (False Positive) is the number of pixels that are incorrectly classified as positive in the predicted mask but not in the ground truth mask.
- FN - (False Negative) is the number of pixels that are incorrectly classified as negative in the predicted mask but are positive in the ground truth mask.

F1 Score:

F1 Score combines precision and recall to assess the balance between false positives and false negatives in the segmentation results. It is a harmonic mean of precision and recall, providing a balance between the two metrics. Precision measures the proportion of true positive predictions out of all positive predictions made by the model, while recall measures the proportion of true positive predictions out of all actual positive instances in the data. The F1 score ranges from 0 to 1, with 1 being the best possible score, indicating perfect precision and recall. A higher F1 score indicates better model performance in terms of both precision and recall.

The F1 score is calculated as follows:

$$F_1 = 2 \cdot \frac{\text{precision} \cdot \text{recall}}{\text{precision} + \text{recall}} \quad (1)$$

$$\text{precision} = \frac{TP}{TP + FP} \quad (2)$$

$$\text{recall} = \frac{TP}{TP + FN} \quad (3)$$

- TP - (True Positive) is the number of correctly classified positive instances.
- FP - (False Positive) is the number of instances that are incorrectly classified as positive.
- FN - (False Negative) is the number of instances that are incorrectly classified as negative.

Other Parameters:

n_classes=1: Indicates the number of classes in the segmentation task. Since this is a binary segmentation task (prostate vs. background), there is only one class.

activation='sigmoid': Specifies the activation function used in the output layer of the model. Sigmoid activation is suitable for binary classification tasks.

epoch = 50: Defines the number of training epochs, i.e., the number of complete passes through the training dataset. Increasing the number of epochs allows the model to learn more complex patterns but may lead to overfitting if not regularized properly.

BATCH_SIZE = 10: Determines the number of samples processed in each training iteration before updating the model's weights. A smaller batch size requires less memory but may result in slower convergence, while a larger batch size can speed up training but may lead to less stable updates.

Experimentation:

We tested the performance of training models on different Learning rates and noticed that Site A performed better at the learning rate higher than required for Site B with Early stopping. Site A field Strength of 3T therefore has higher Resolution whereas the Site B has field strength of 1.5T and has lower Resolution that is why it requires smaller Learning rate to perform better result on test set.

VGG19	LR	epochs	Loss	mean iou_score	mean f1-score
SITE A	0.0001	50	0.1323	0.8152	0.89302
	0.001	50	0.15659	0.77234	0.86222
	0.001	34	0.13674	0.79001	0.87815
	0.01	32	0.42922	0.48266	0.62561
SITE B	0.0001	50	0.29092	0.62448	0.74278
	0.0001	16	0.42561	0.47391	0.60686
	0.001	22	0.51238	0.43207	0.56744
	0.01	17	0.8517	0.15996	0.2621

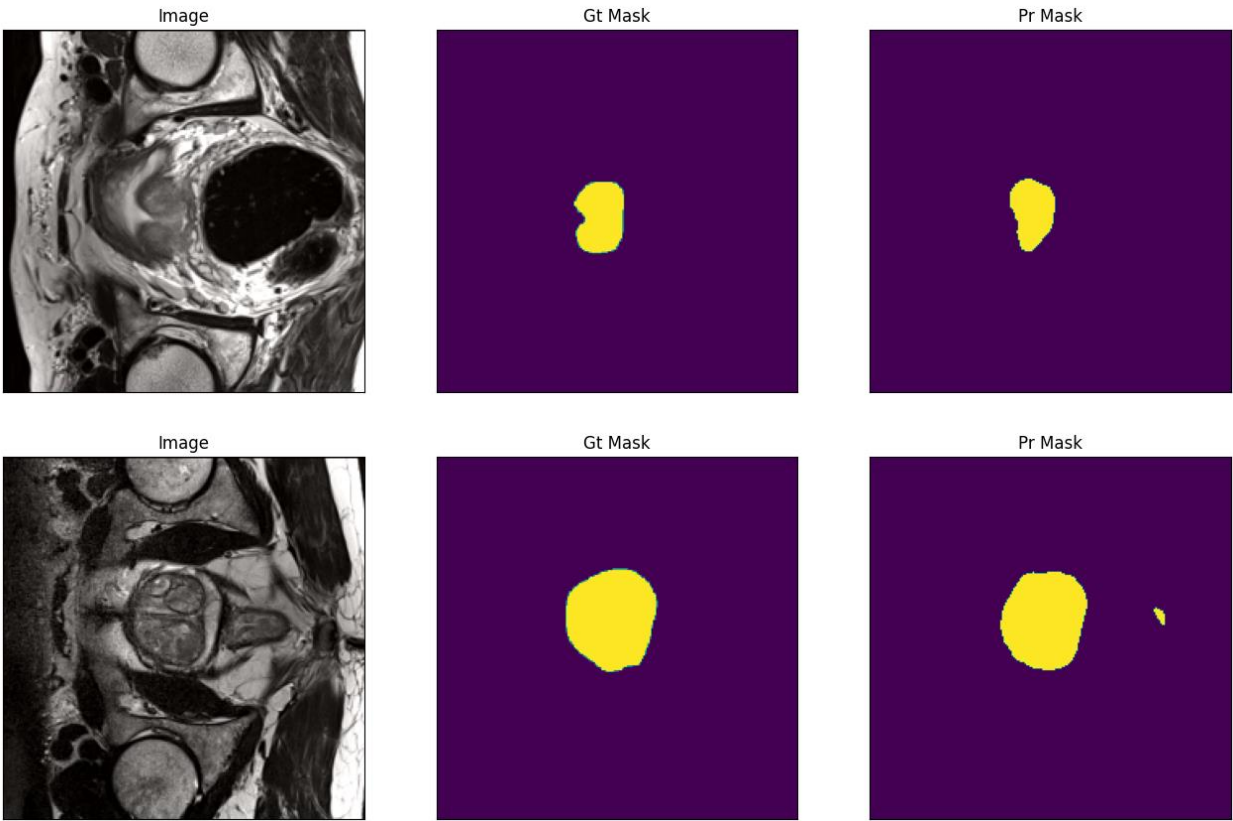
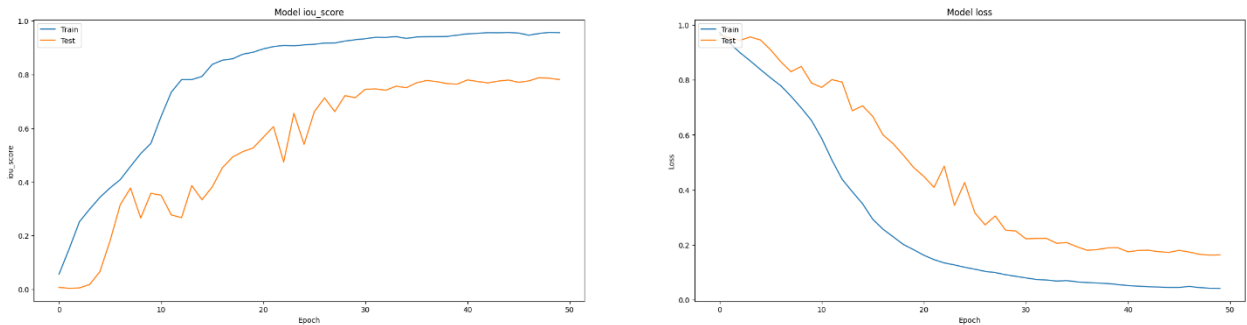
RESNET50	LR	epochs	Loss	mean iou_score	mean f1-score
SITE A	0.0001	50	0.14889	0.84024	0.91032
	0.001	9	0.89538	0.19022	0.25064
	0.001	34	0.13569	0.79542	0.88085
SITE B	0.001	31	0.15887	0.7752	0.85734
	0.001	34	0.13829	0.79763	0.87781
	0.001	6	0.81414	0.43038	0.58981
	0.0001	34	0.15887	0.7752	0.85734
	0.001	13	0.21704	0.73127	0.81857

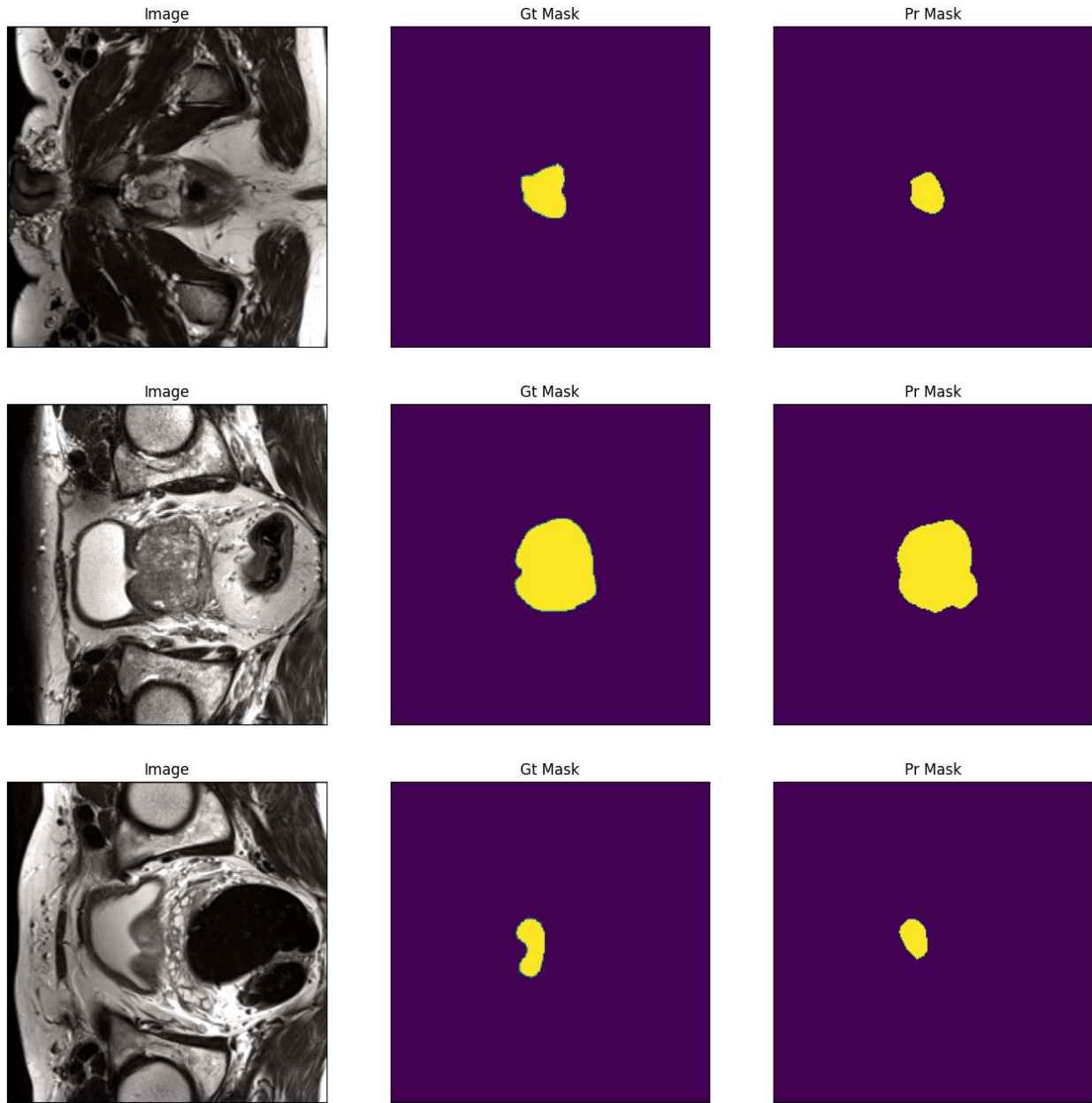
EFFICIENTNETB0	LR	Epochs	Loss	mean iou_score	mean f1-score
SITE A	0.001	13	0.10565	0.84557	0.91272
	0.001	5	1.3296	0.12996	0.22632
SITE B	0.001	17	0.16874	0.77517	0.85335
	0.0001	34	0.31722	0.71212	0.81236

SITE A

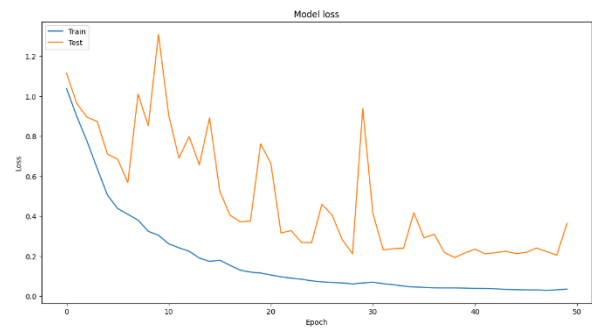
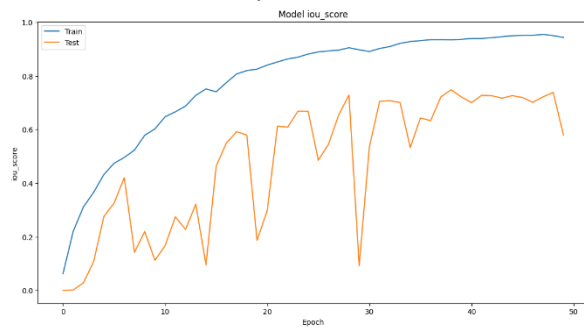
VGG19

VGG19: SITE A – 50 Epochs – LR:0.0001

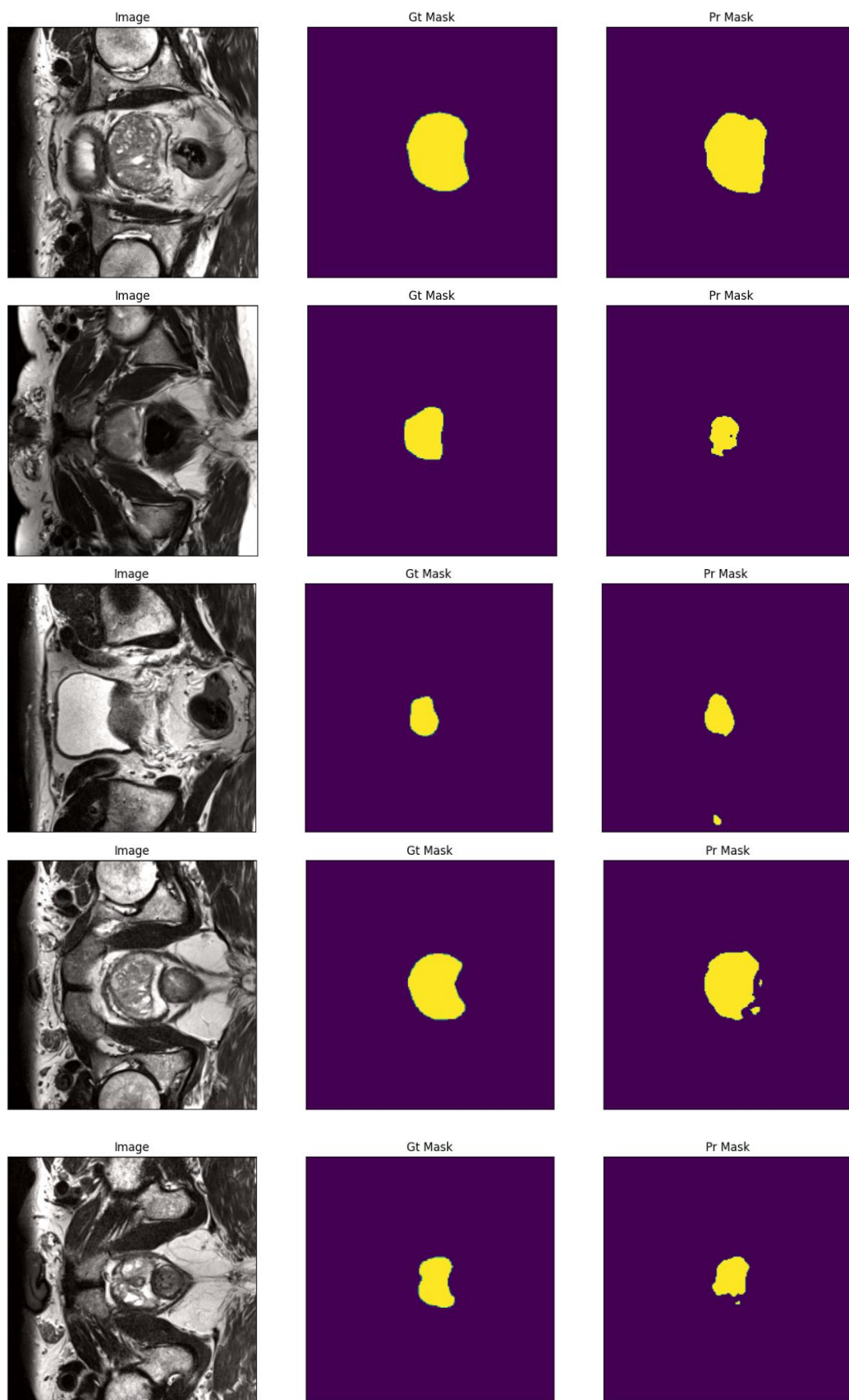




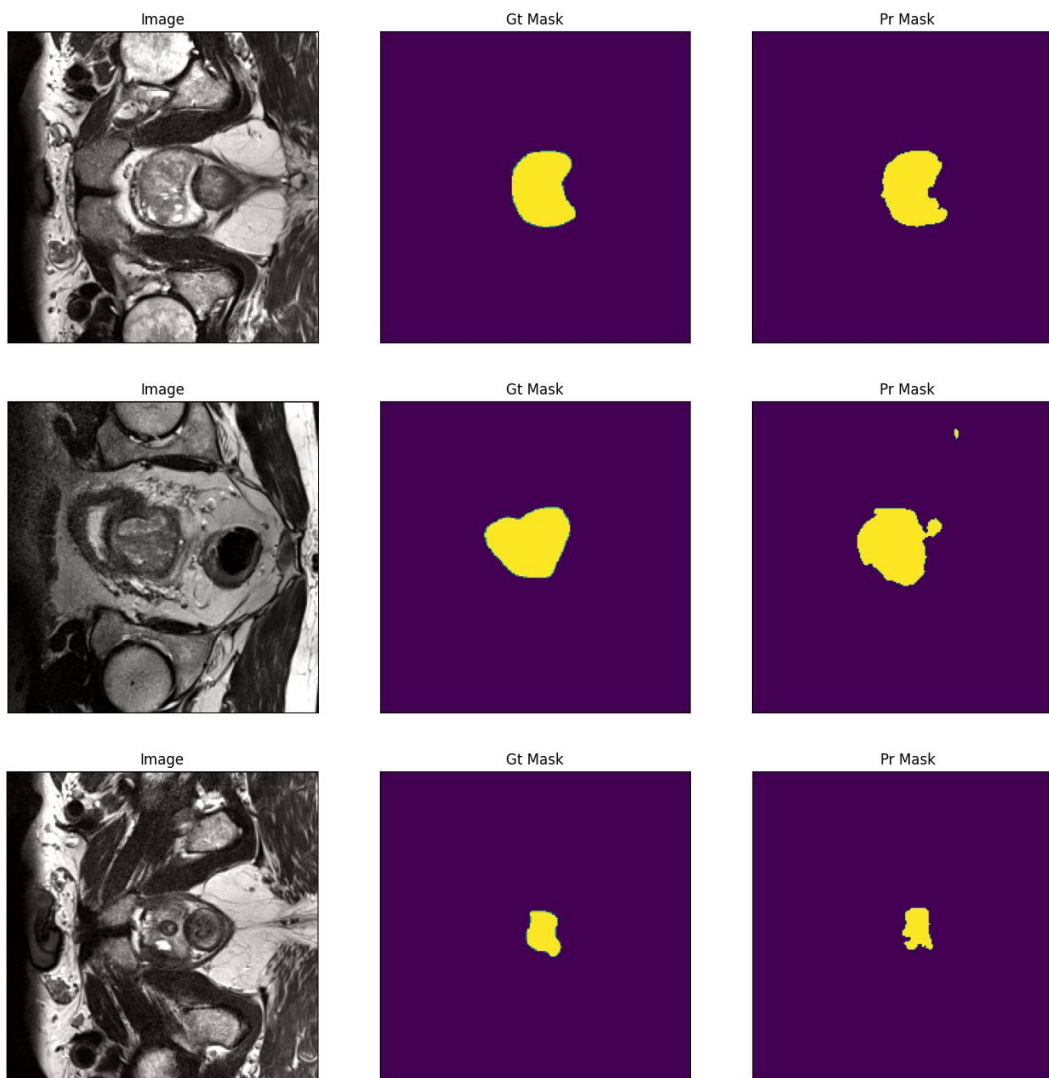
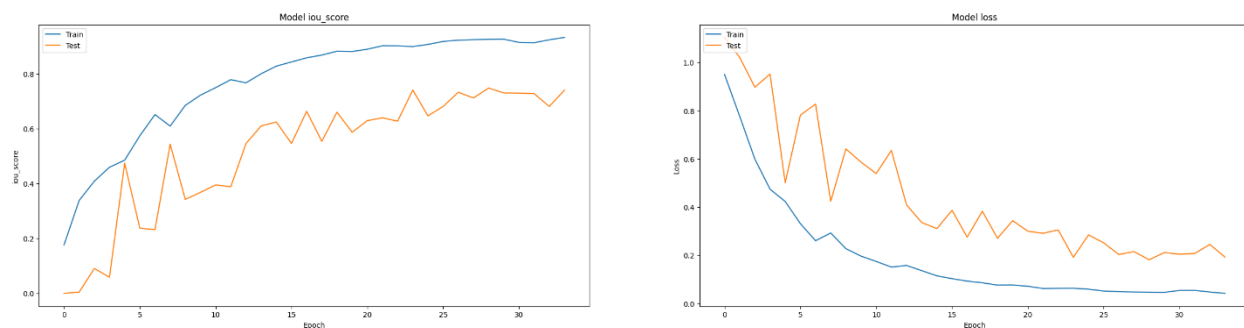
VGG19: SITE A – 50 Epochs – LR:0.001



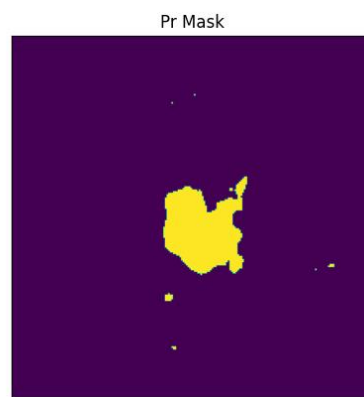
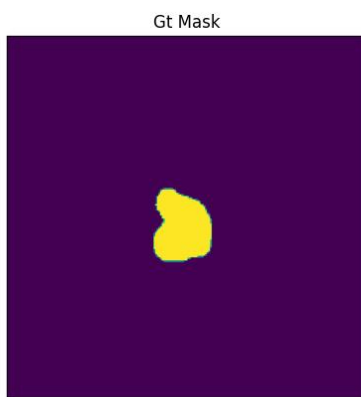
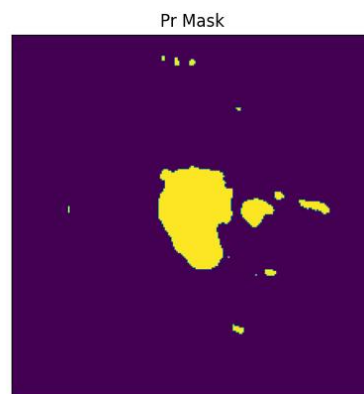
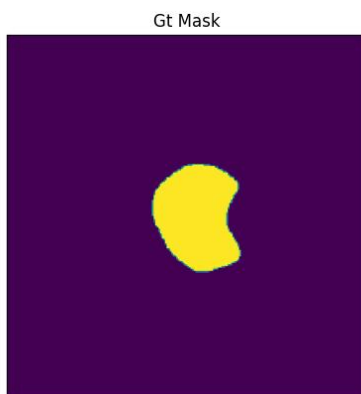
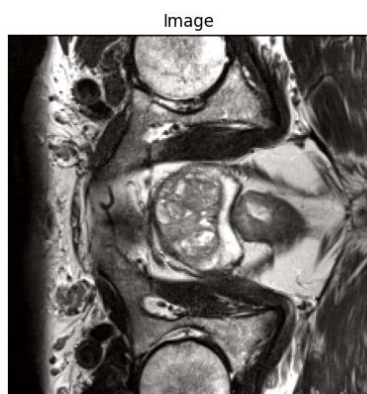
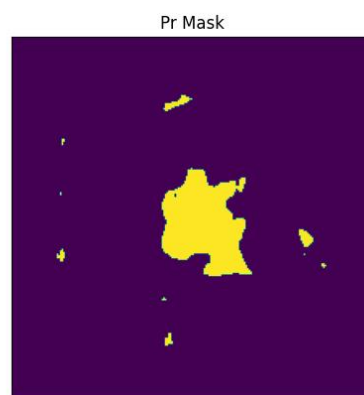
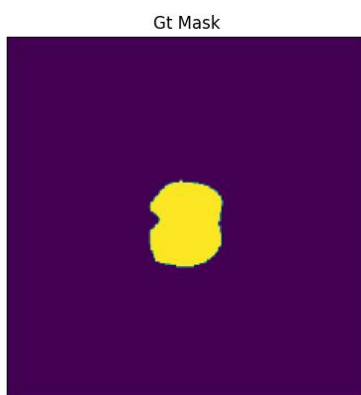
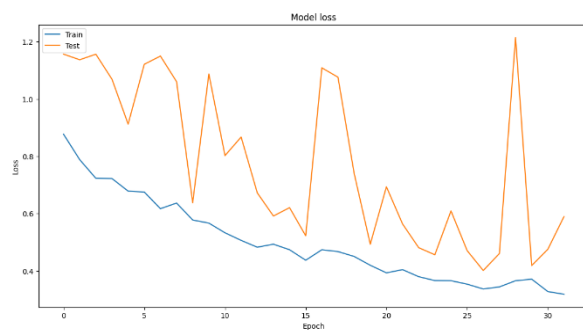
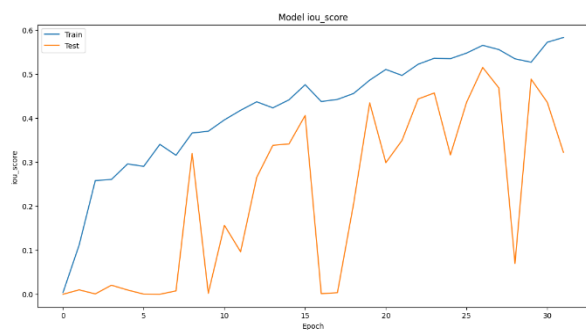
Test Images Results:

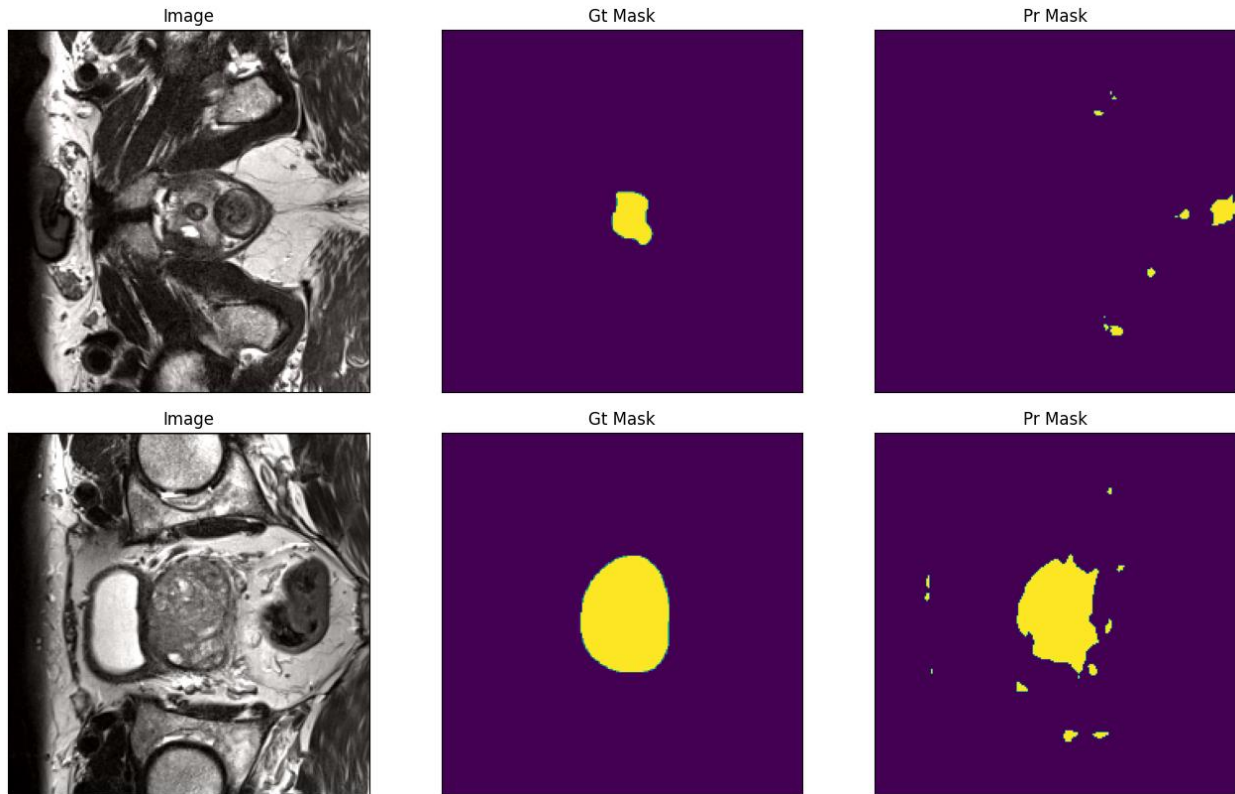


VGG19: SITE A – 50 Epochs – LR:0.001- Early Stopping 34 epochs - Loss: 0.13674 | mean iou_score: 0.79001 |mean f1-score: 0.87815



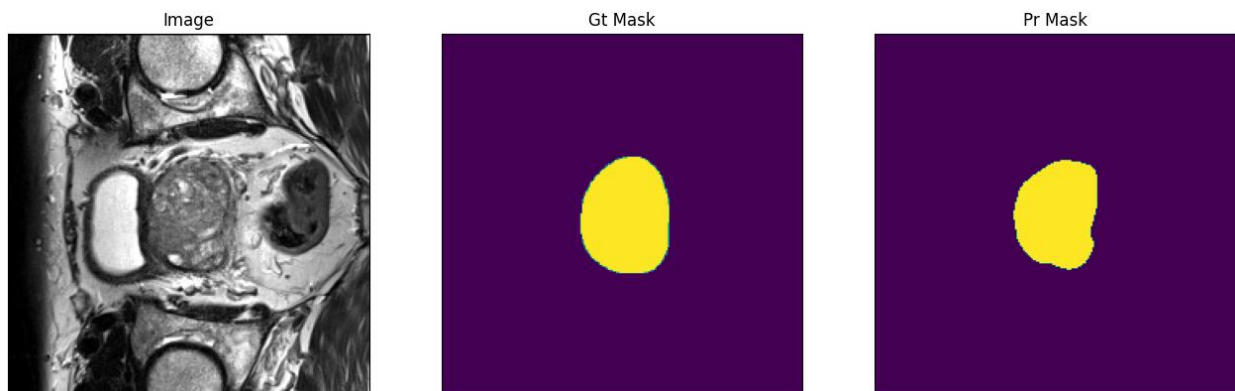
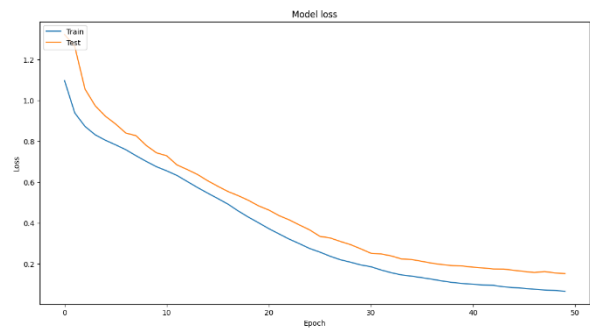
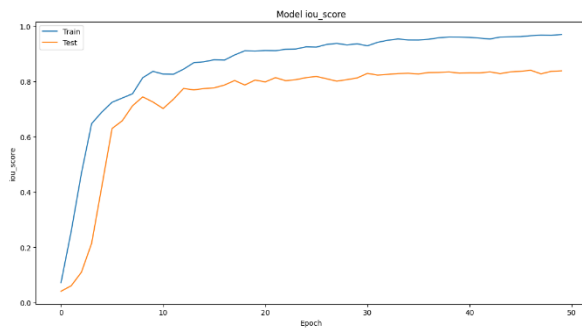
VGG19: SITE A – 50 Epochs – LR:0.01- Early Stopping 32 epochs



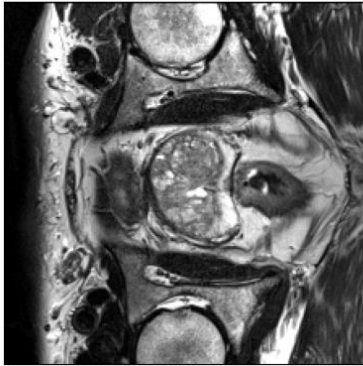


RESNET50

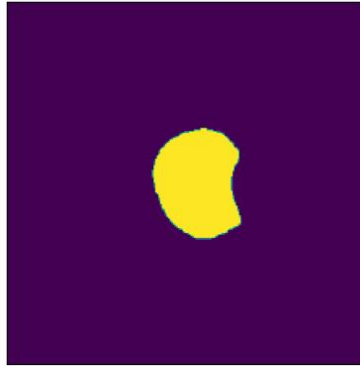
RESNET50: SITE A – 50 Epochs – LR 0.0001 – Loss: 0.14889 | mean iou_score: 0.84024 | mean f1-score: 0.91032



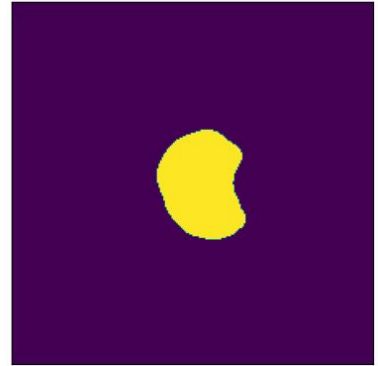
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Gt Mask



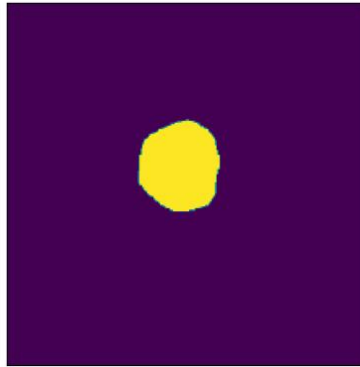
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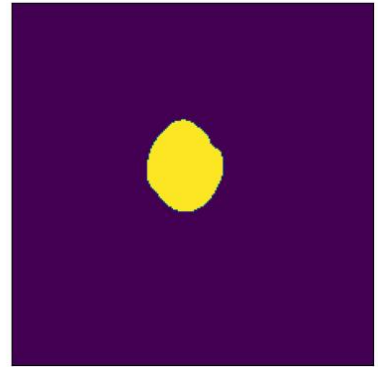
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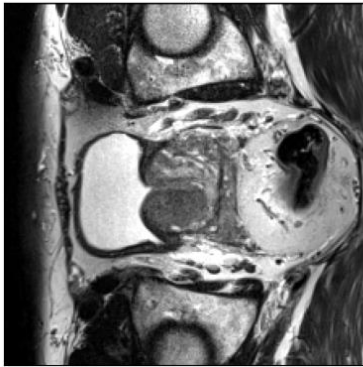
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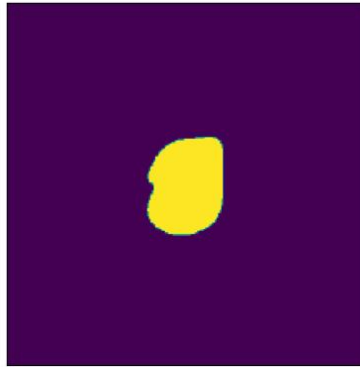
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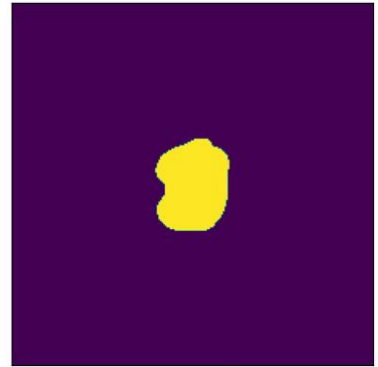
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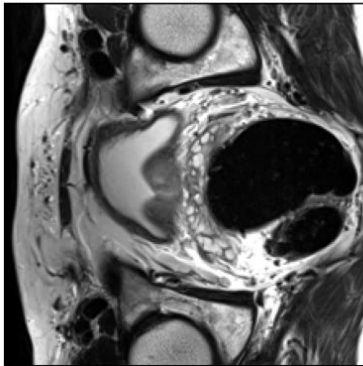
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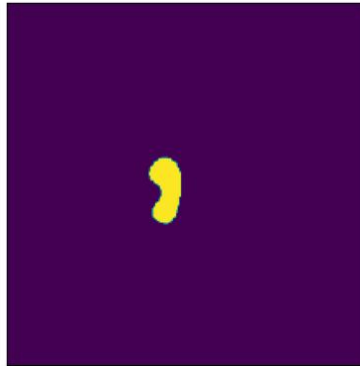
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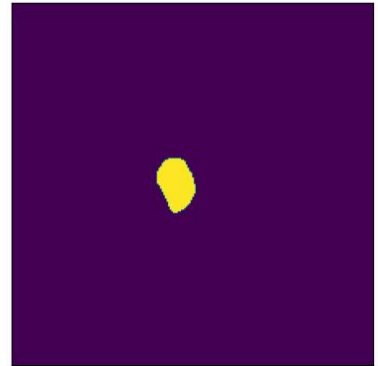
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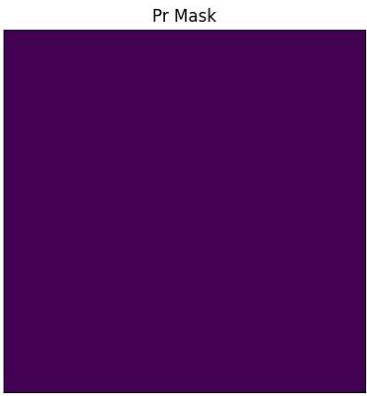
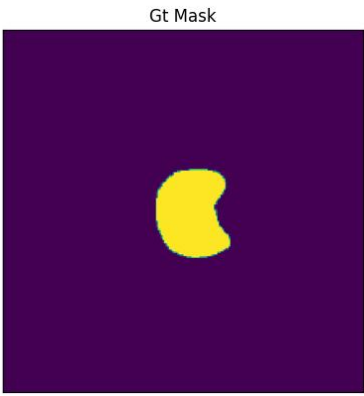
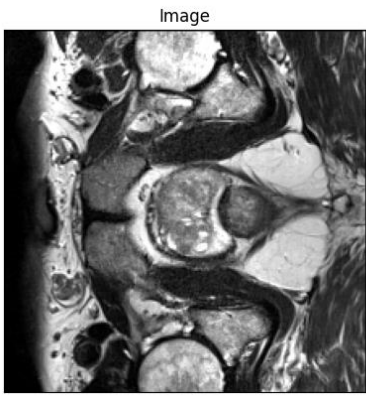
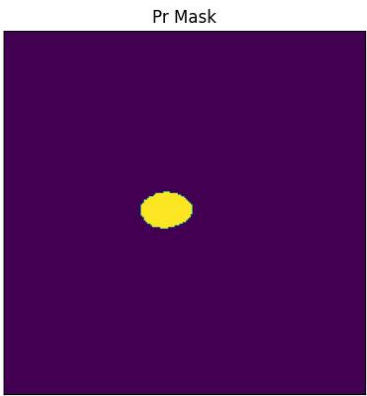
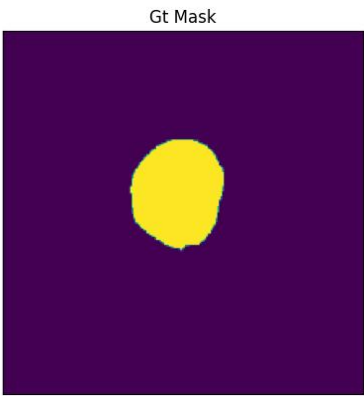
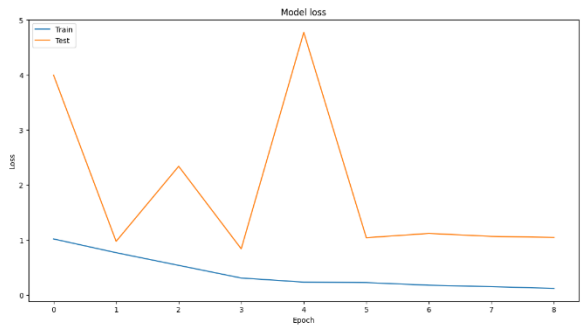
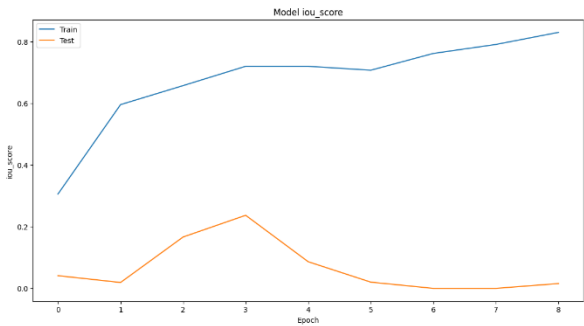
Gt Mask



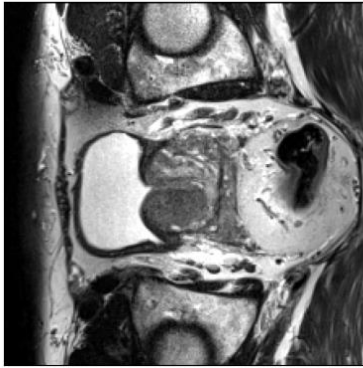
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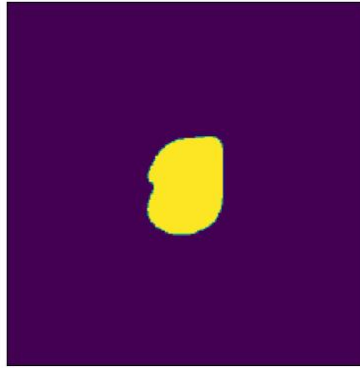
RESNET50: SITE A – 50 Epochs – Early Stopping: 9 Epoch - LR 0.001 – Loss: 0.89538 | mean iou_score: 0.19022 | mean f1-score: 0.25064



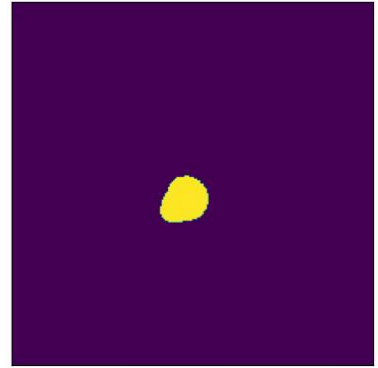
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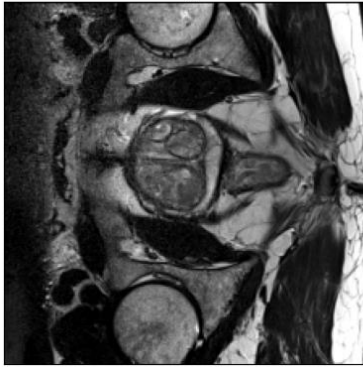
Gt Mask



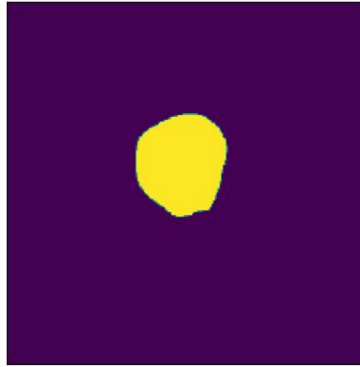
Pr Mask



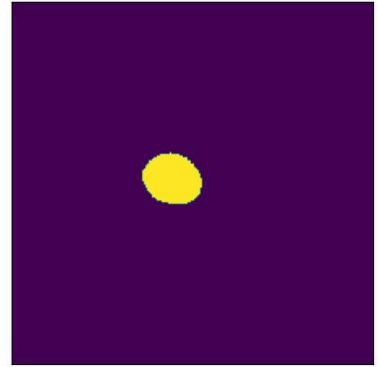
Image



Gt Mask



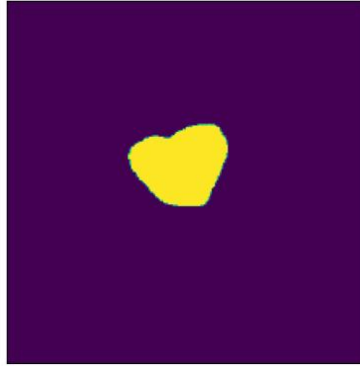
Pr Mask



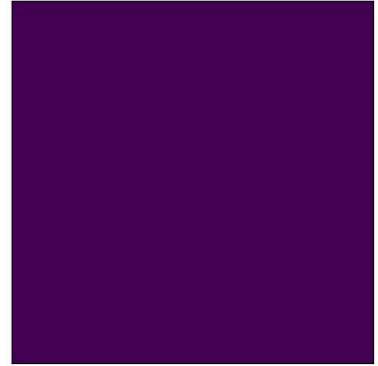
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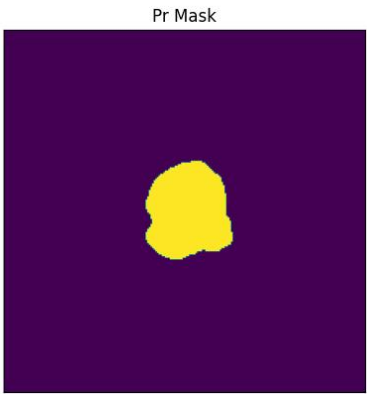
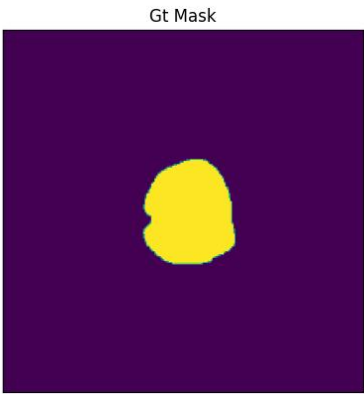
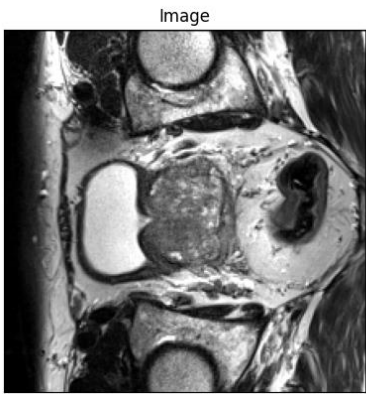
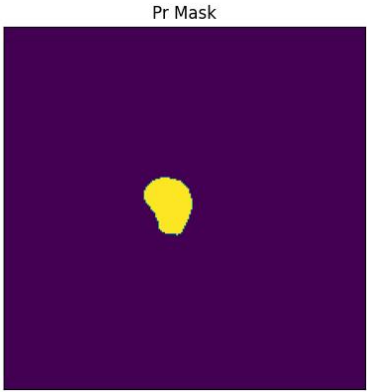
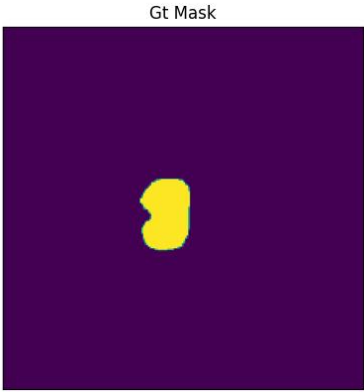
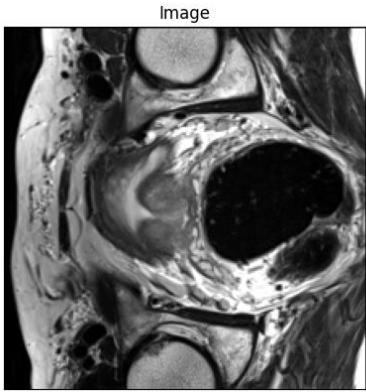
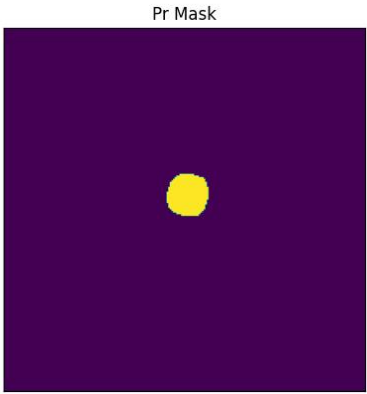
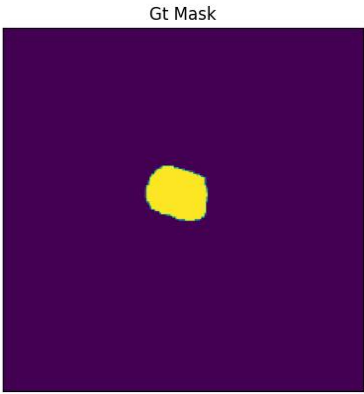
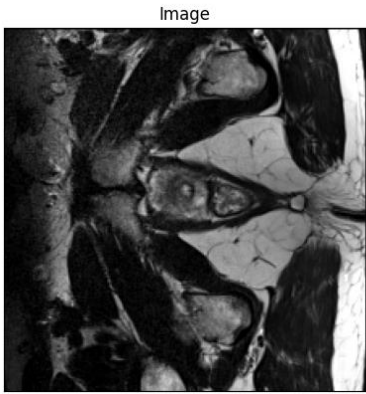
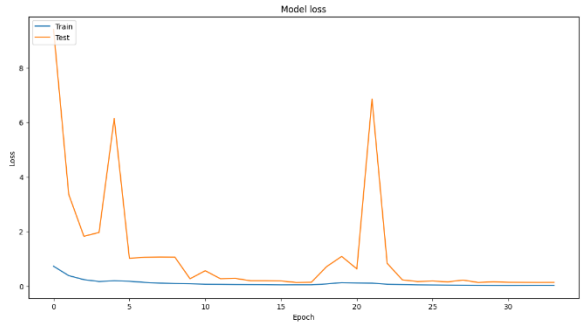
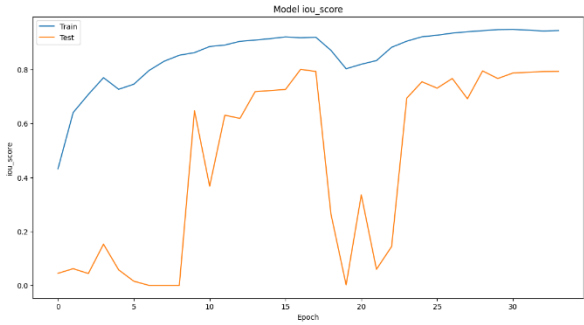
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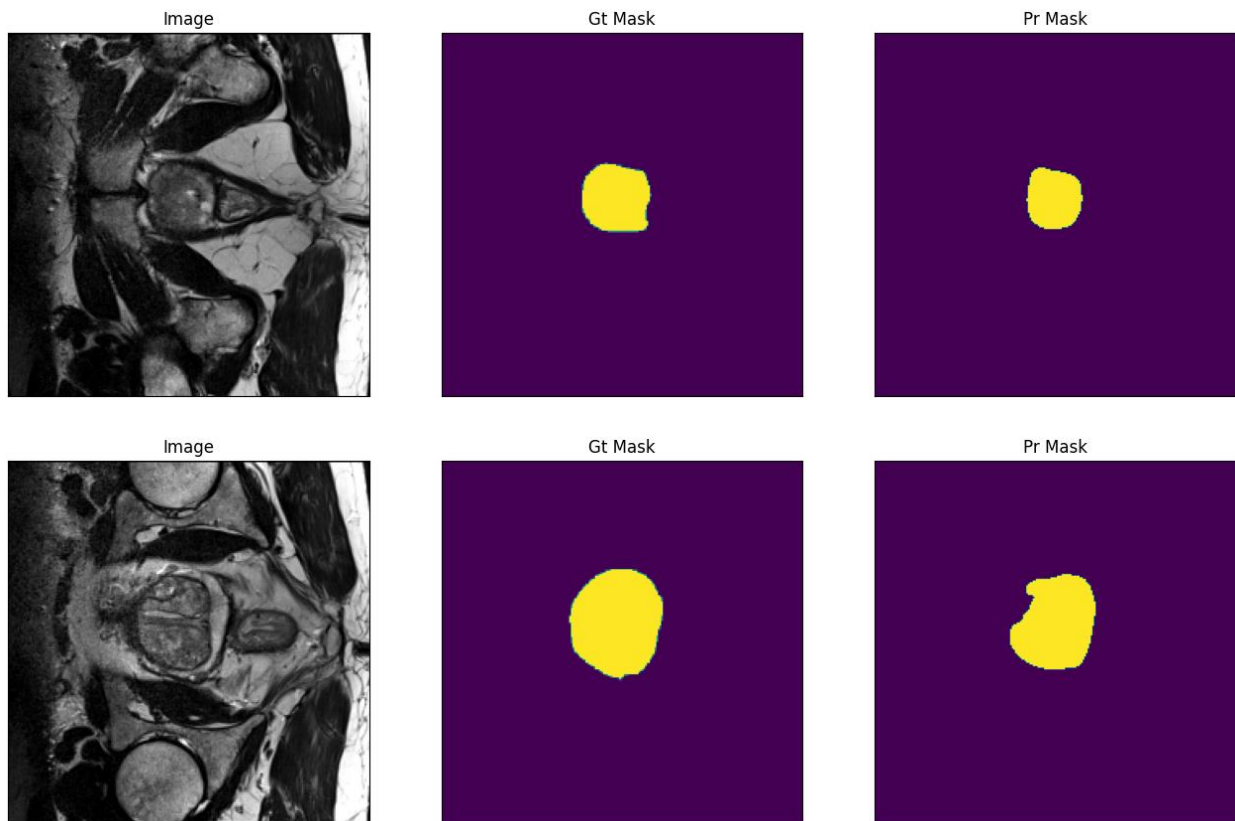


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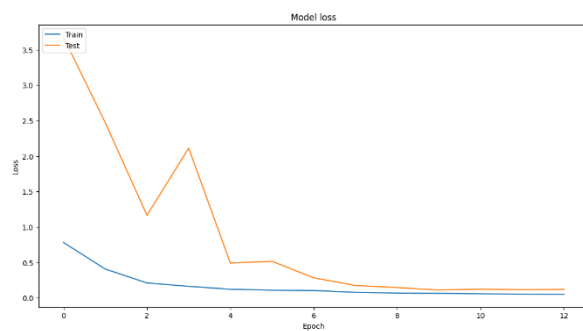
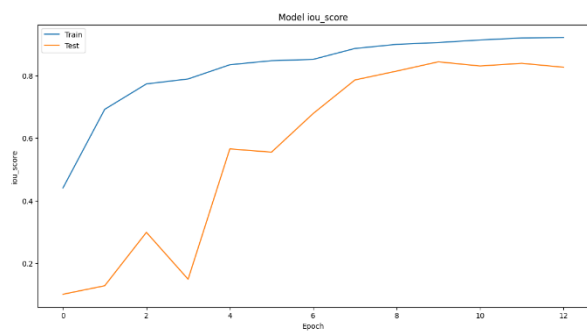
RESNET50: SITE A – 34 Epochs - LR 0.001 – Loss: 0.13569 | mean iou_score: 0.79542 | mean f1-score: 0.88085



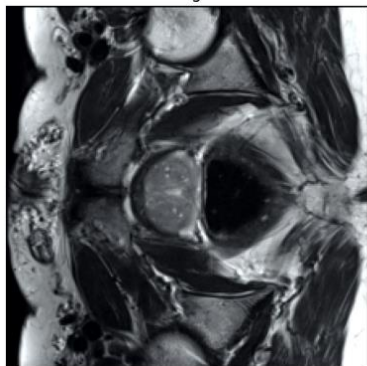


EFFICIENTNETB0

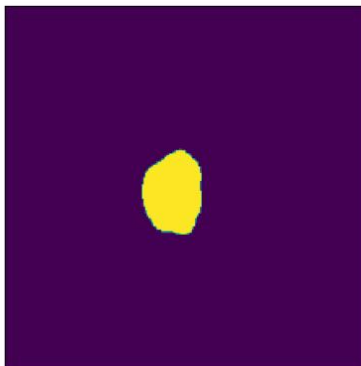
EFFICIENTNETB0: SITE A – Epochs 13 – LR 0.001 – Loss: 0.10565 | mean iou_score: 0.84557 | mean f1-score: 0.91272



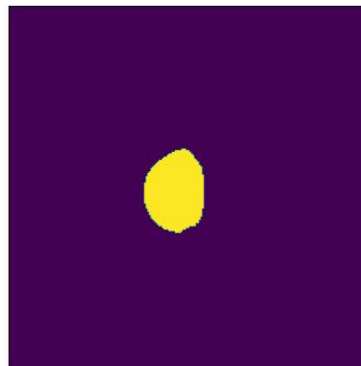
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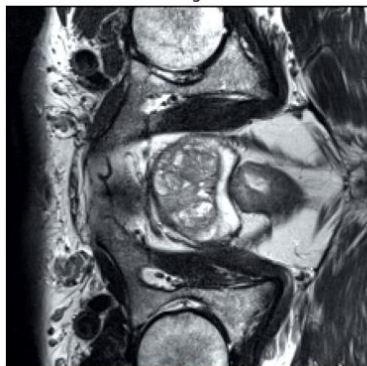
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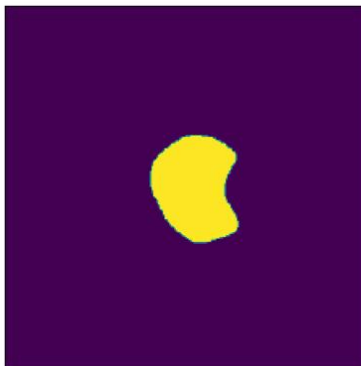
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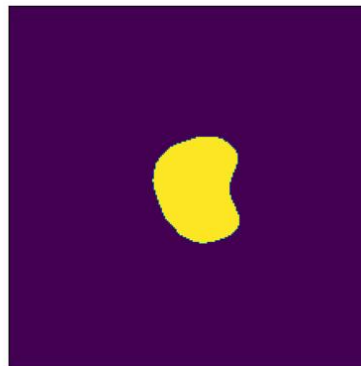
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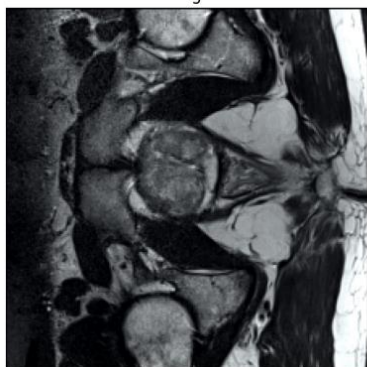
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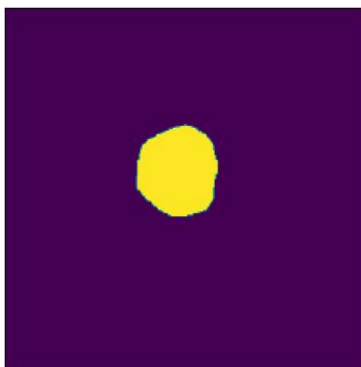
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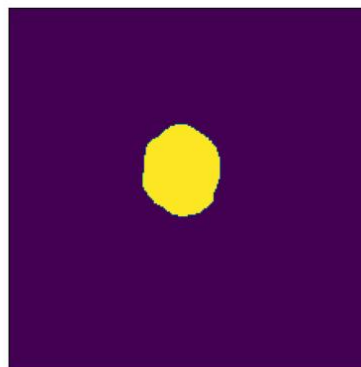
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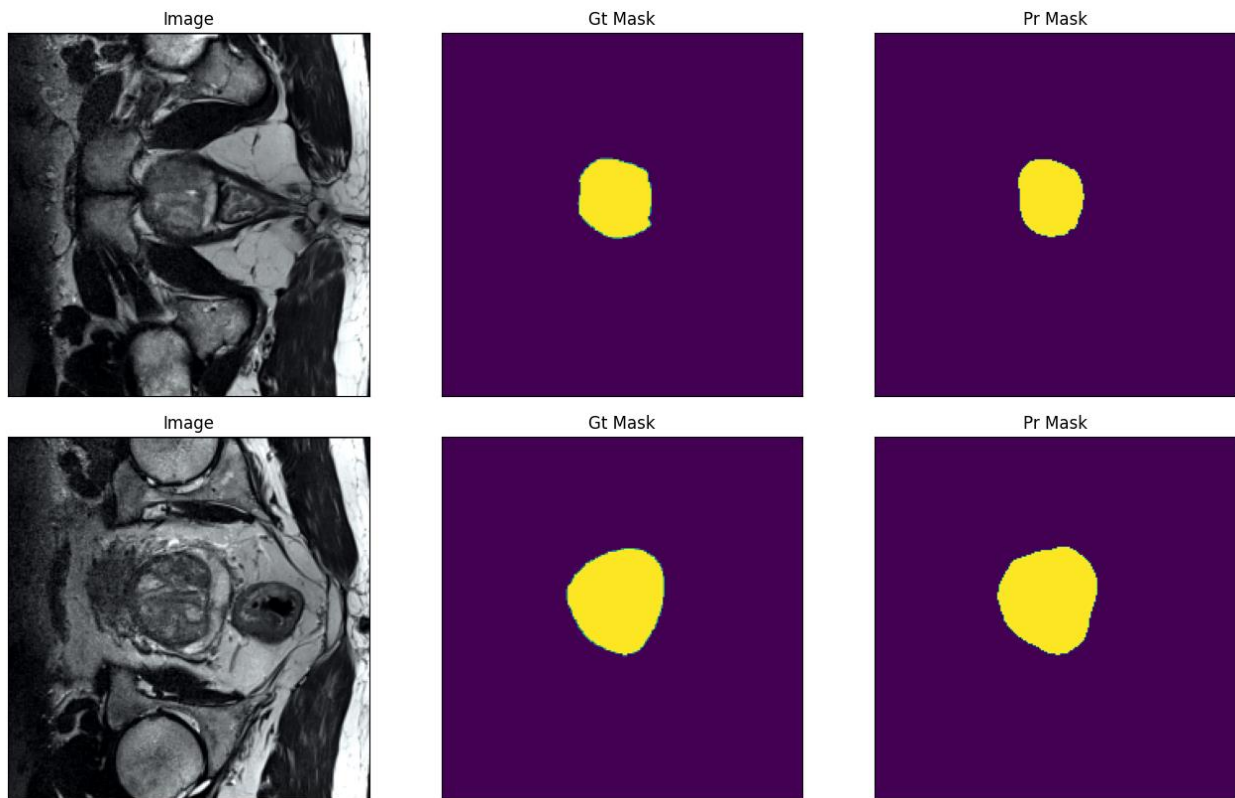


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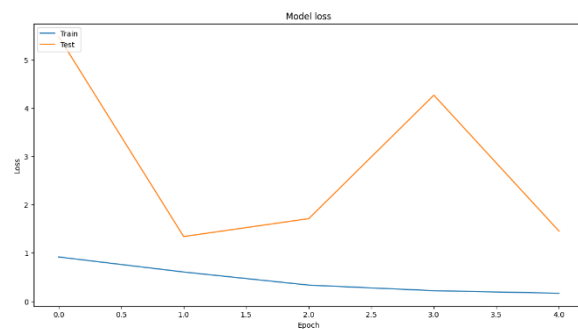
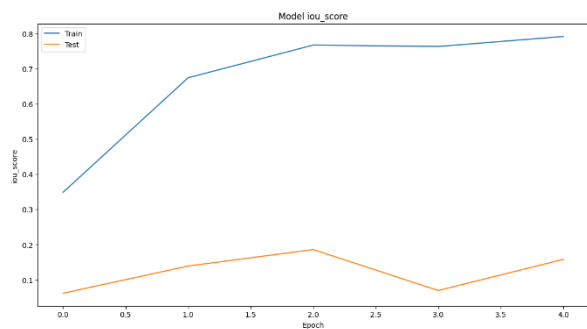


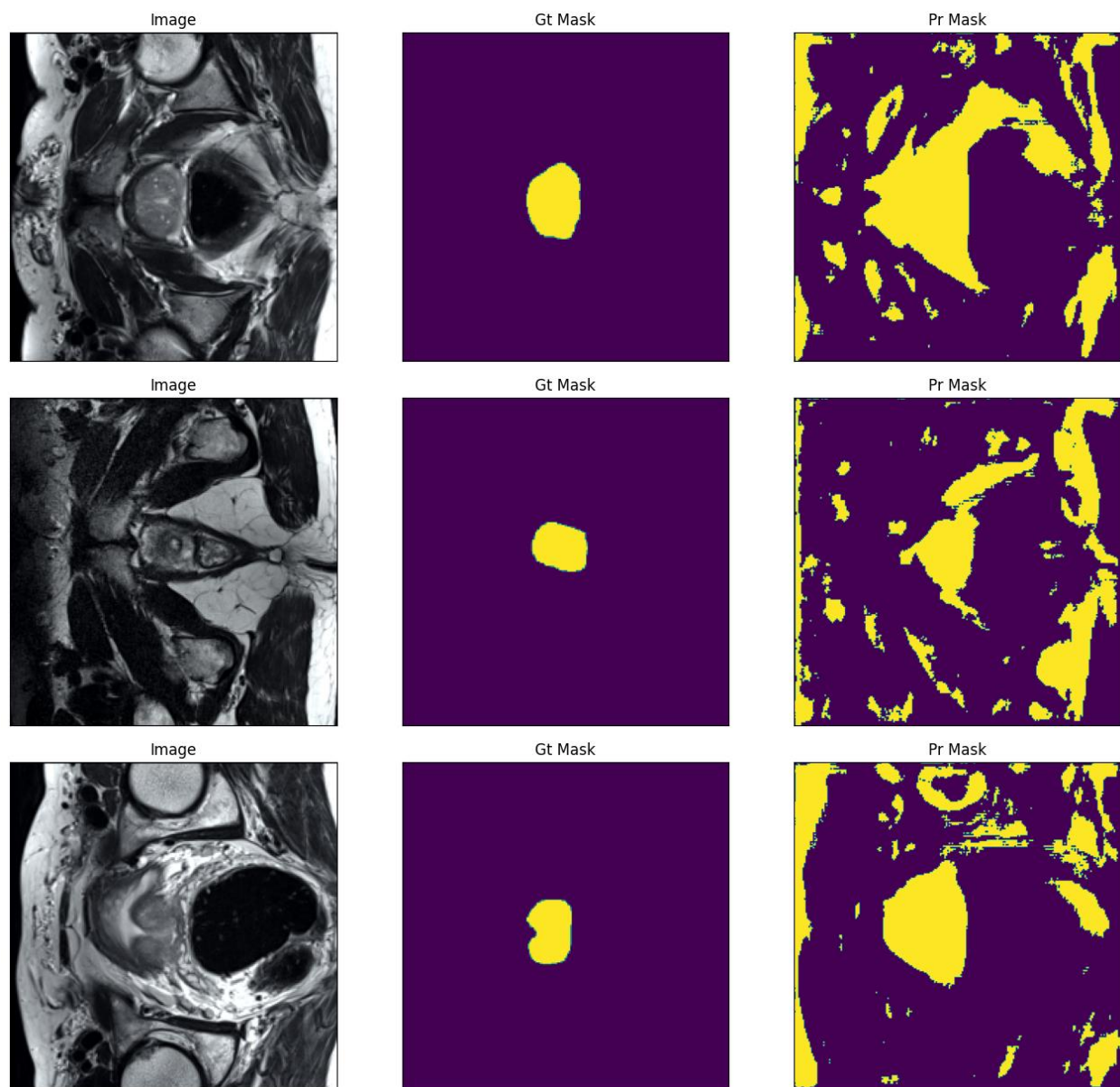
Pr Mask





EFFICIENTNETB0: SITE A – Epochs 5 – LR 0.001 – Loss: 1.3296 | mean iou_score: 0.12996 | mean f1-score: 0.22632

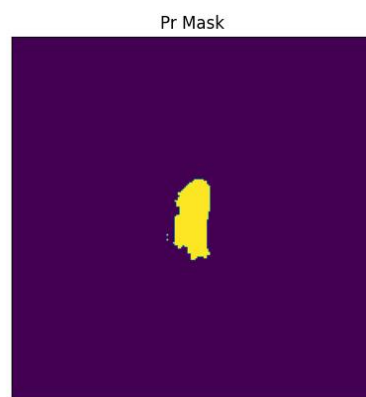
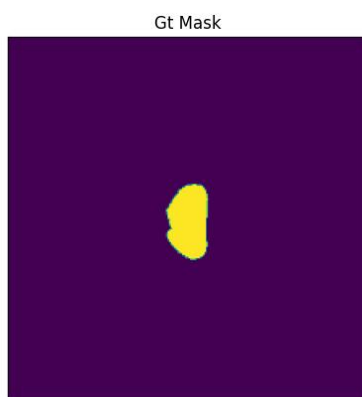
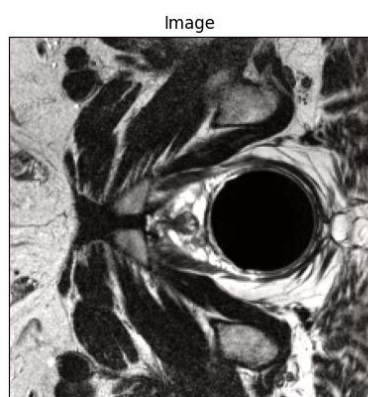
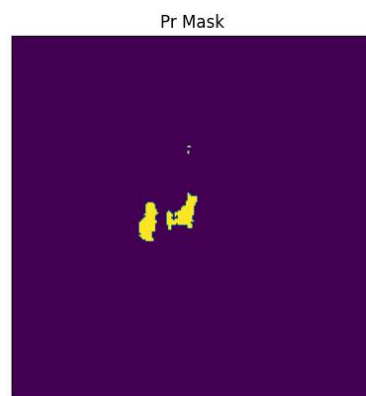
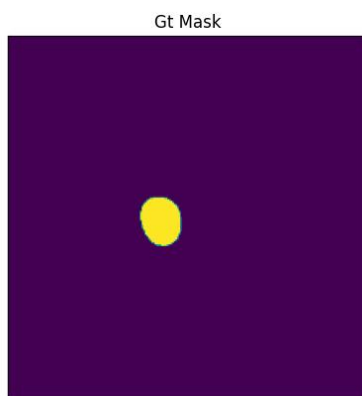
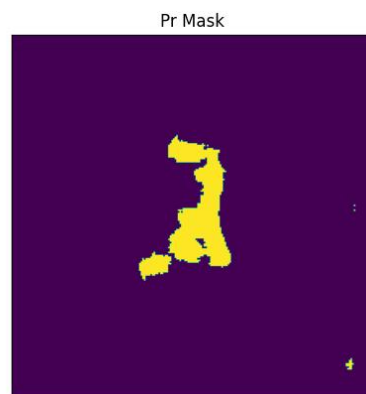
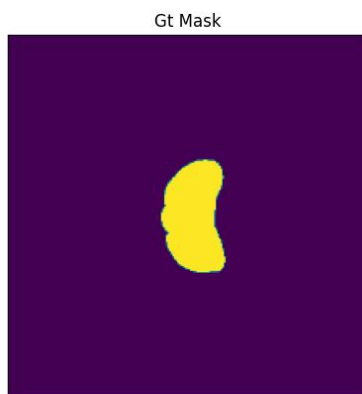
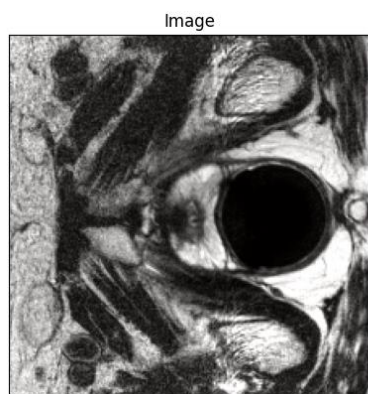
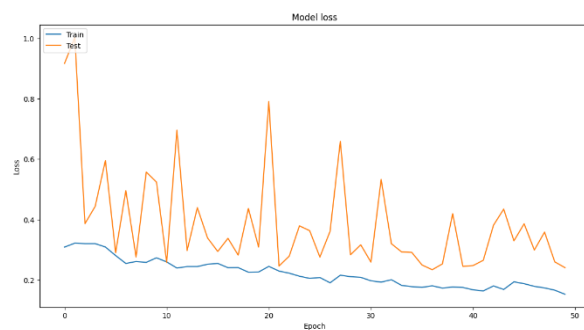
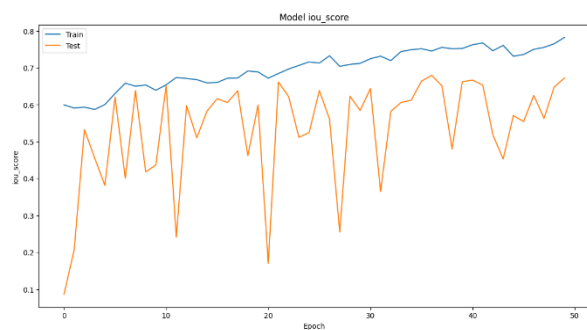


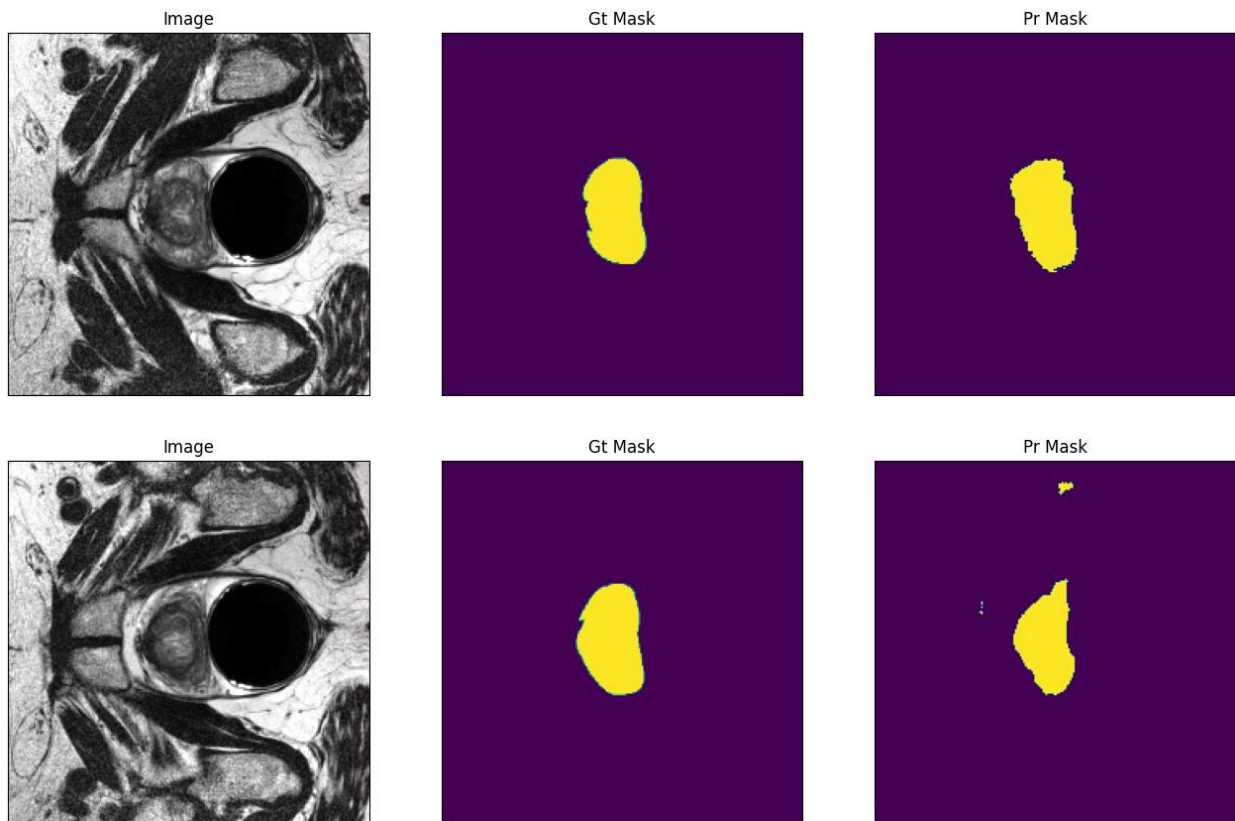


SITE B

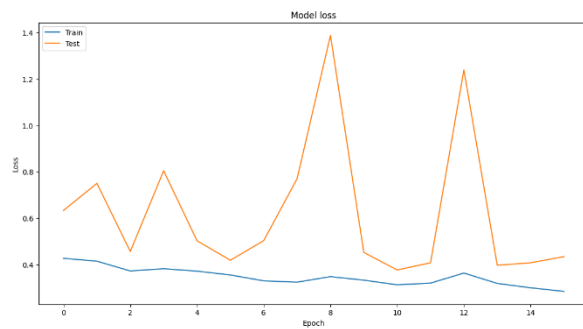
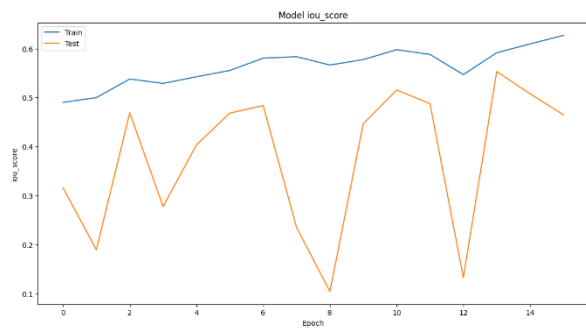
VGG19

VGG19: SITE B – 50 Epochs – LR 0.0001 - Loss: 0.29092 | mean iou_score: 0.62448 | mean f1-score: 0.74278

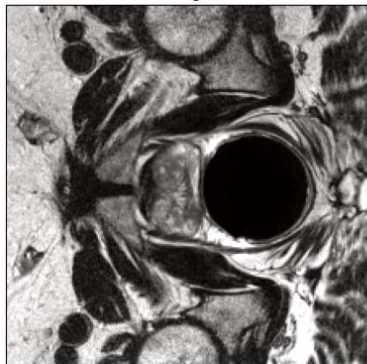




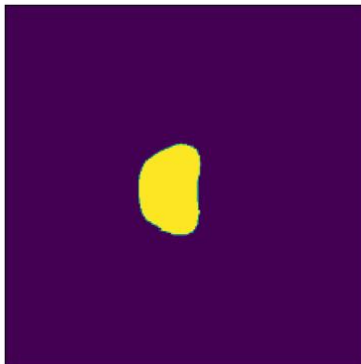
VGG19: SITE B – 50 Epochs – LR 0.0001 - Early Stopping 16 Epoch - Loss: 0.42561 | mean iou_score: 0.47391 | mean f1-score: 0.60686



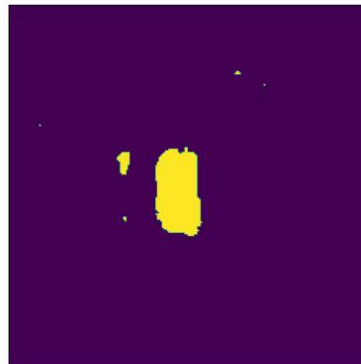
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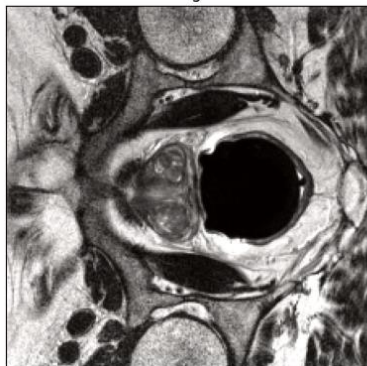
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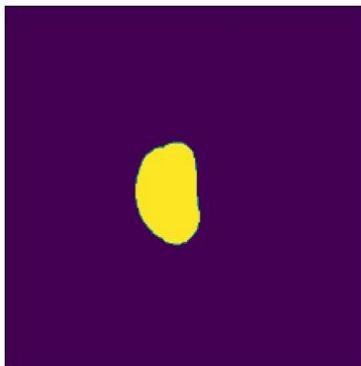
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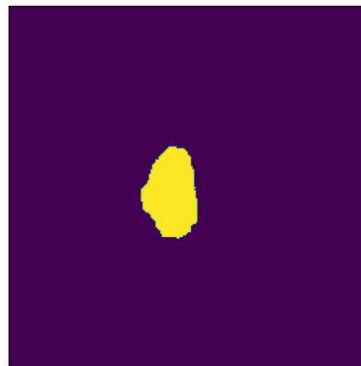
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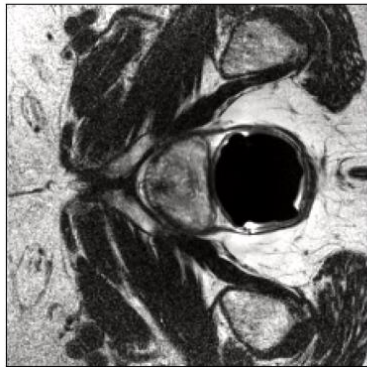
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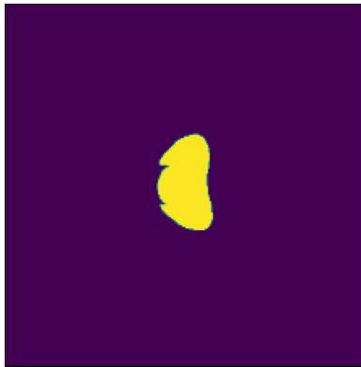
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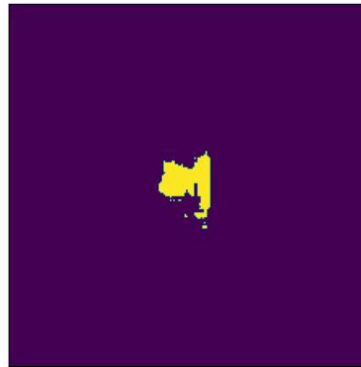
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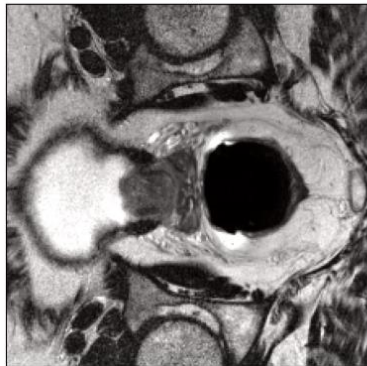
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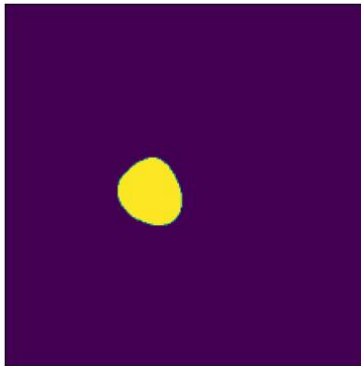
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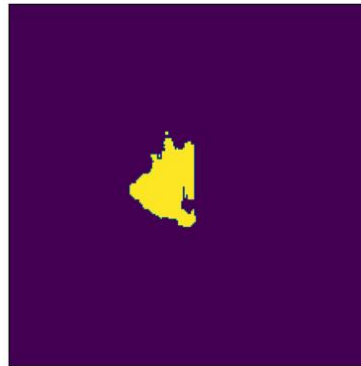
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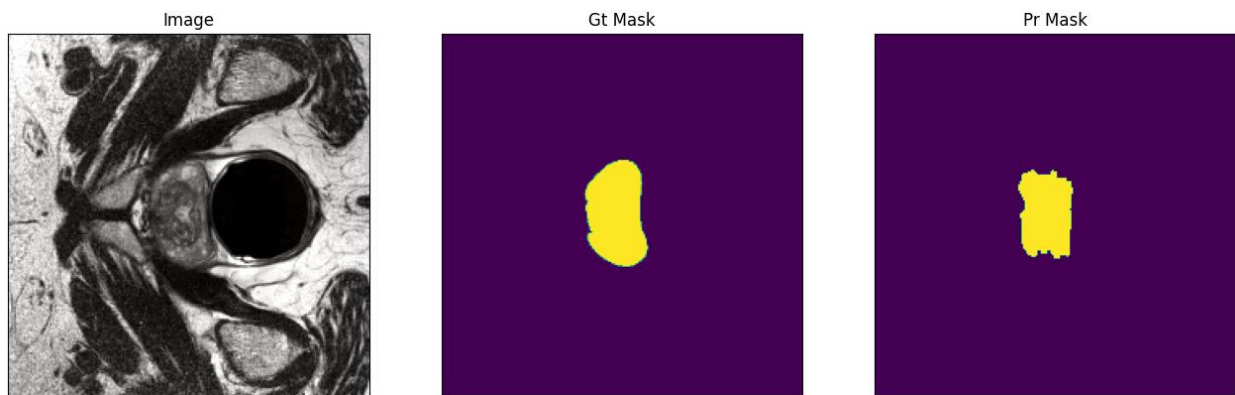


Gt Mask

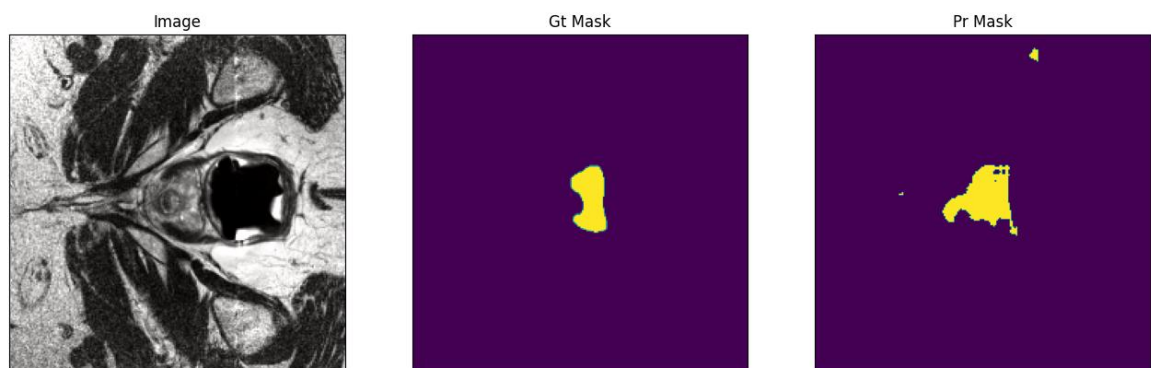
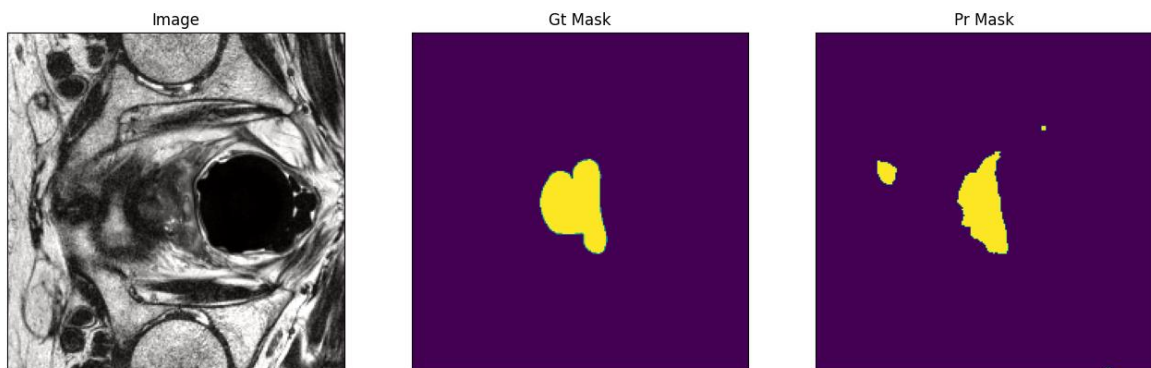
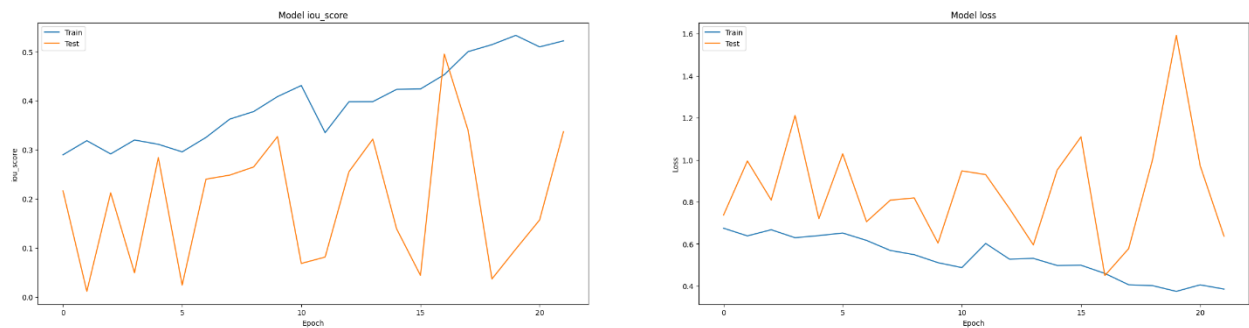


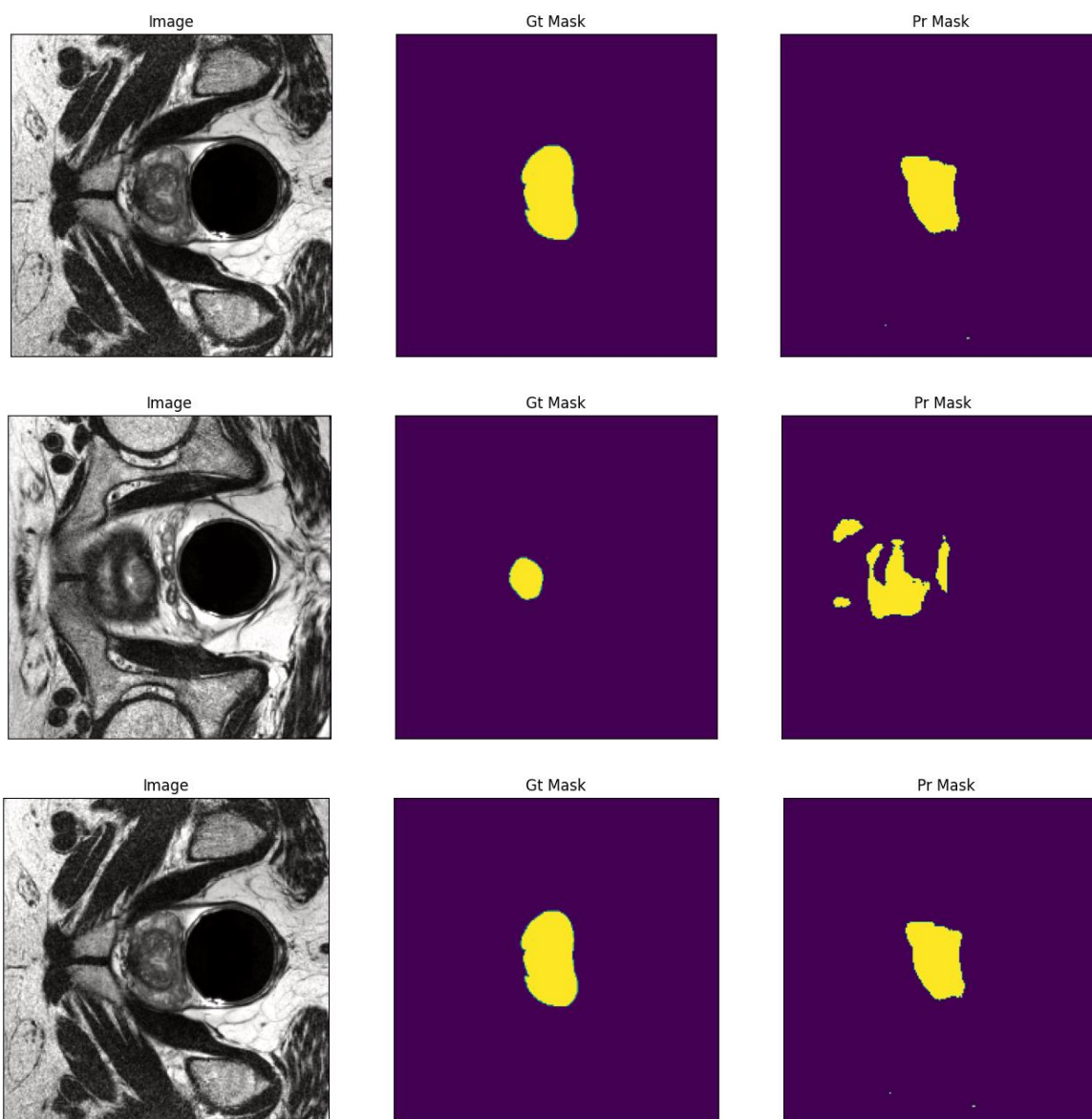
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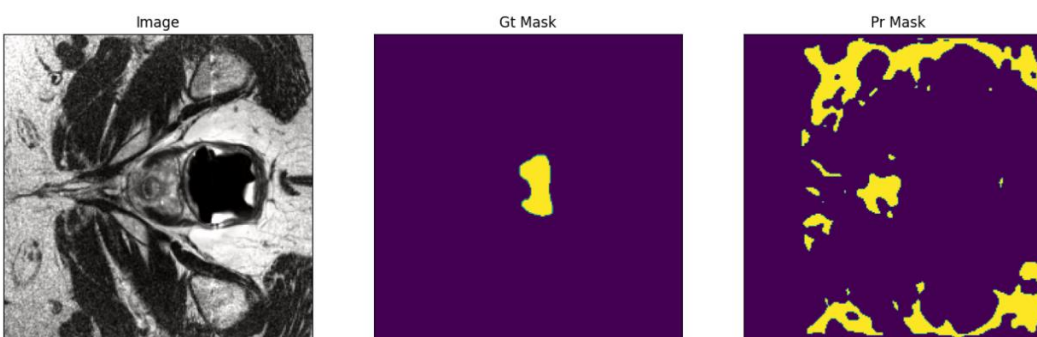
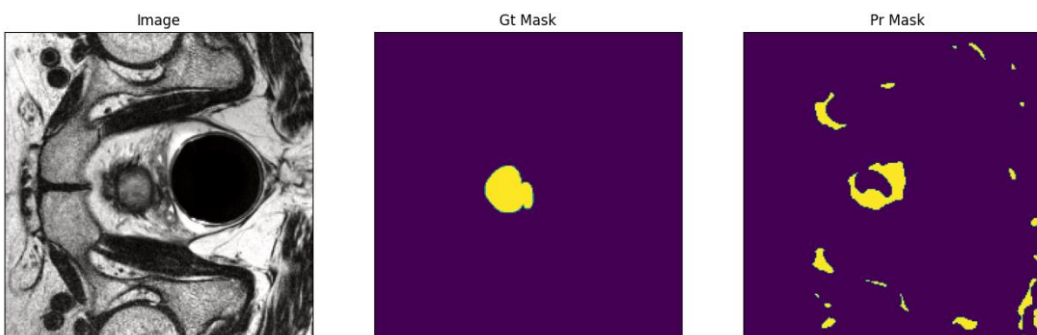
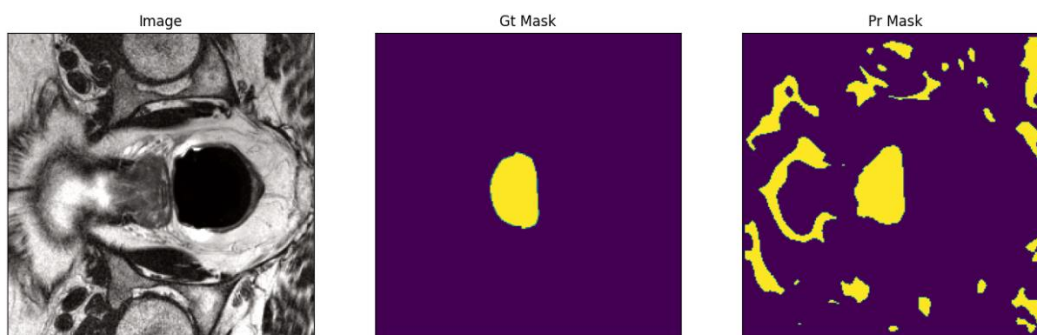
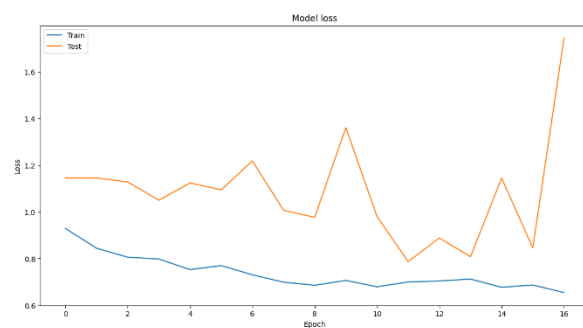
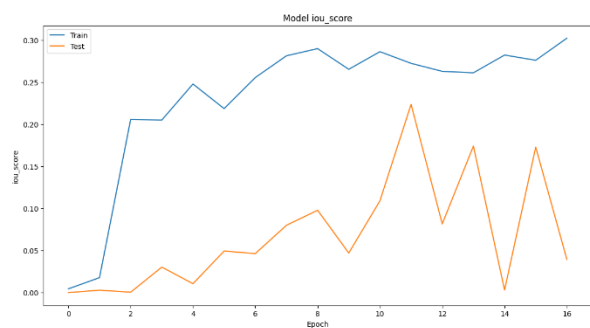


VGG19: SITE B – 50 Epochs – LR 0.001 - Early Stopping 22 Epoch - Loss: 0.51238 | mean iou_score: 0.43207 | mean f1-score: 0.56744



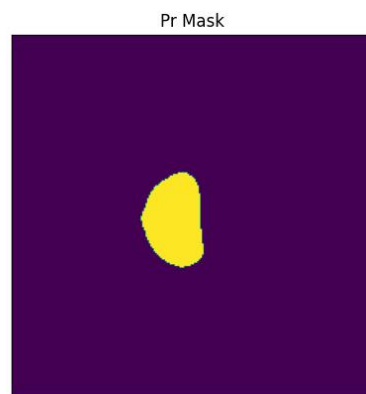
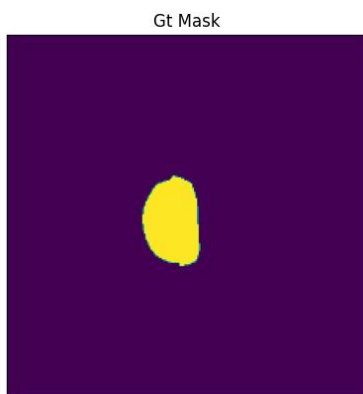
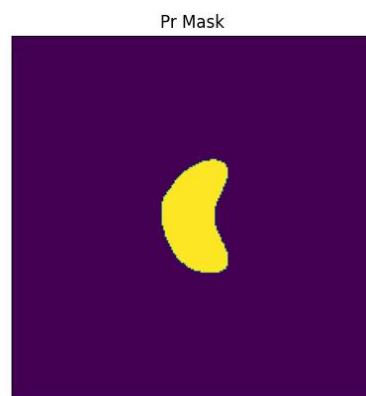
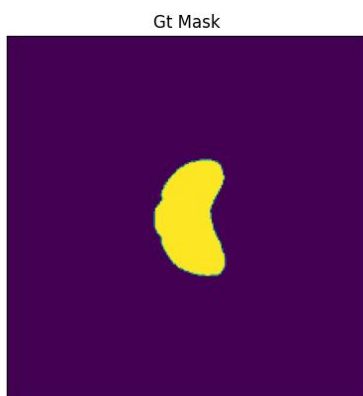
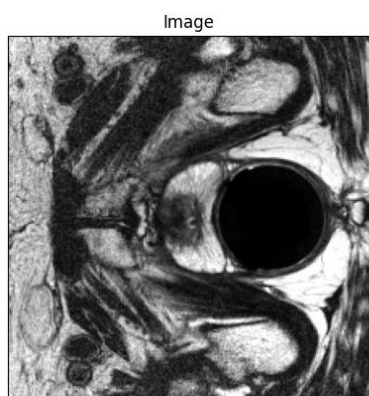
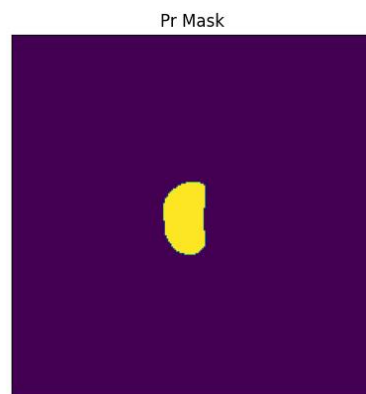
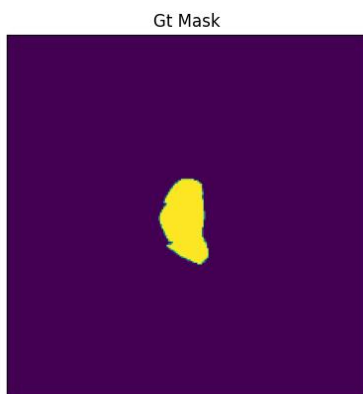
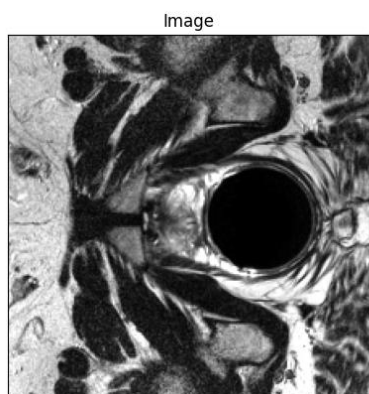
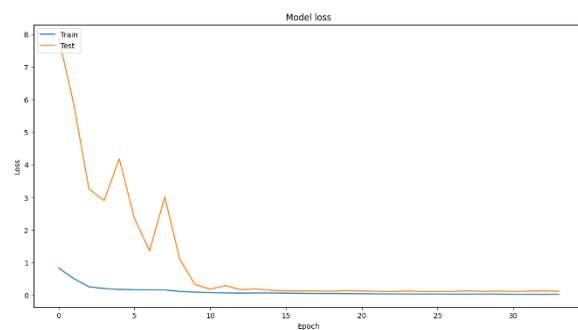
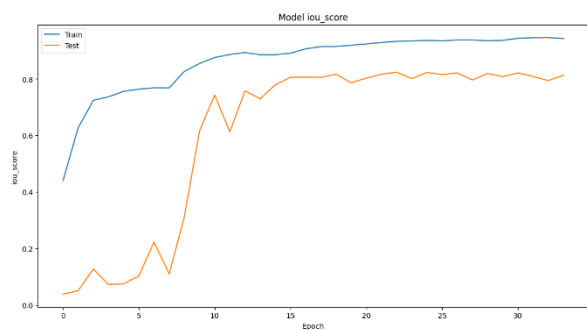


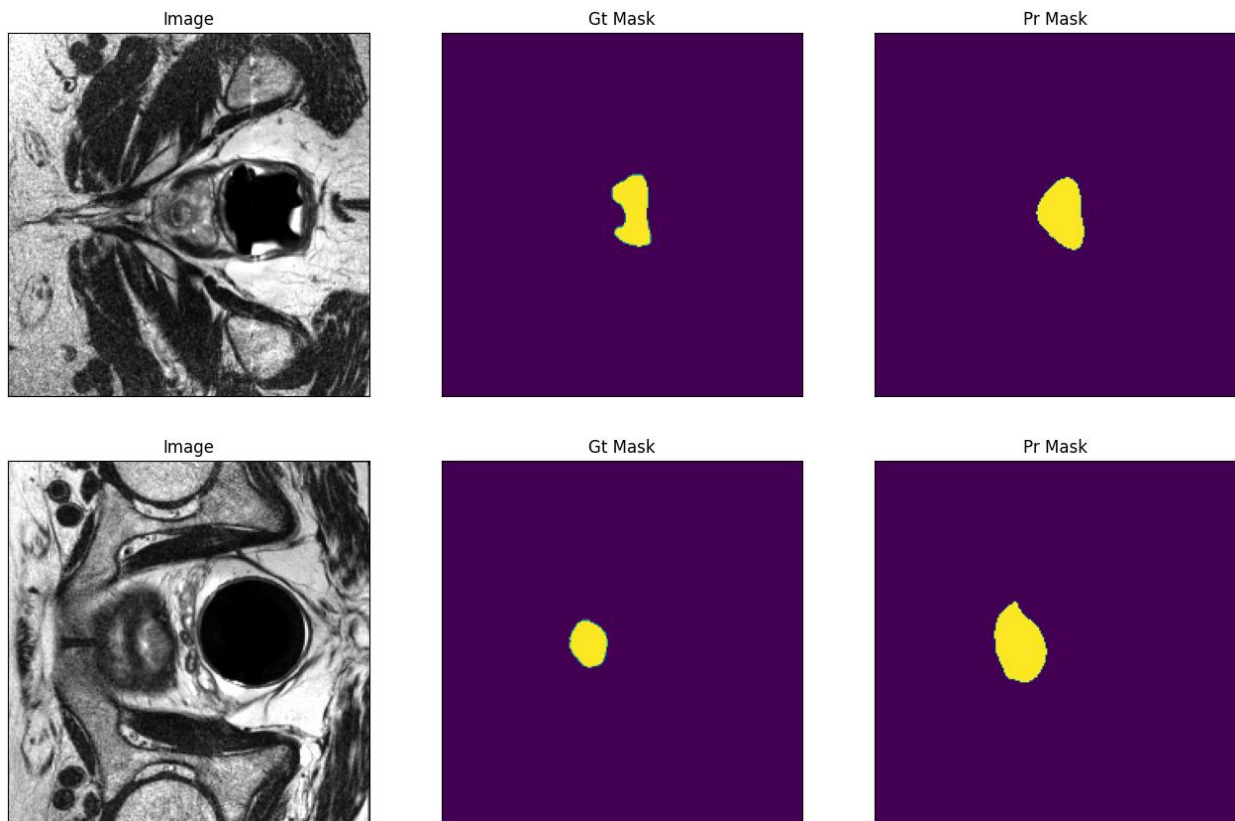
VGG19: SITE B – 50 Epochs – LR 0.01 - Early Stopping 17 Epoch-: Loss: 0.8517 | mean iou_score: 0.15996
| mean f1-score: 0.2621



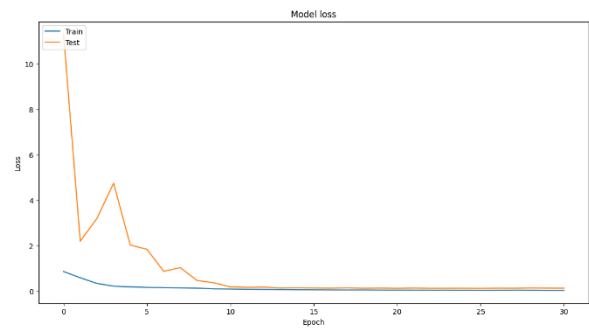
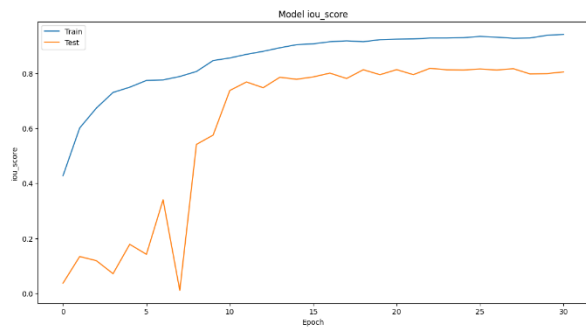
RESNET50

RESNET50: SITE B - 34 Epochs – LR 0.001 – Loss: 0.13829 | mean iou_score: 0.79763 | mean f1-score: 0.87781

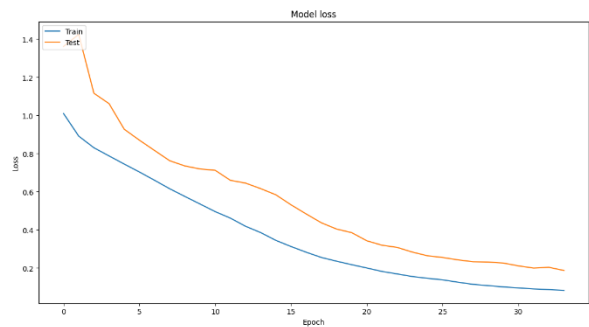
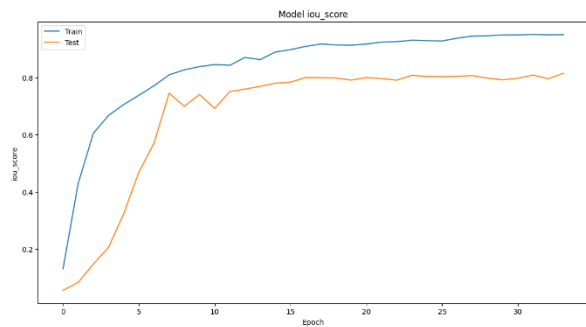




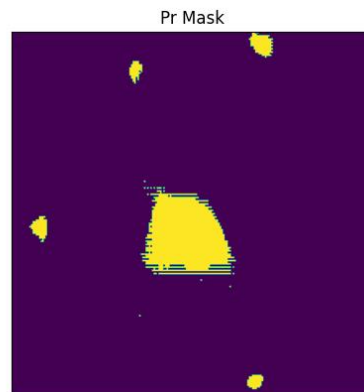
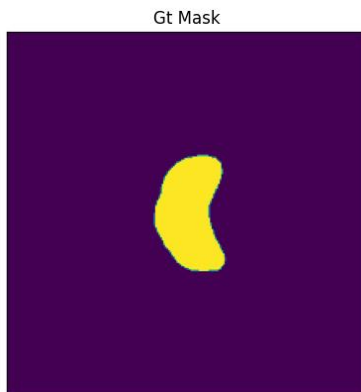
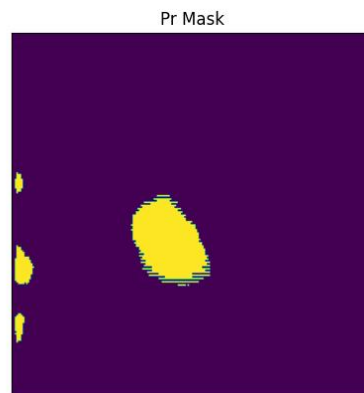
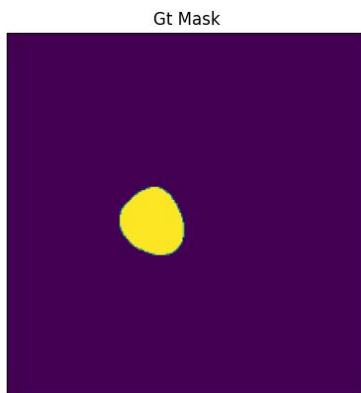
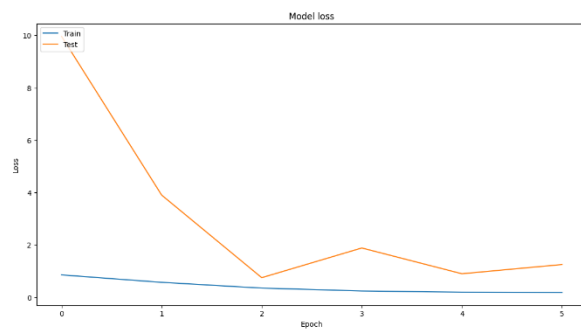
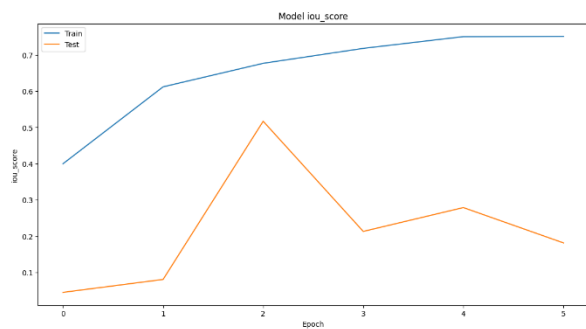
RESNET50: SITE B - 31 Epochs – LR 0.001 – Loss: 0.13829 | mean iou_score: 0.79763 | mean f1-score: 0.87781

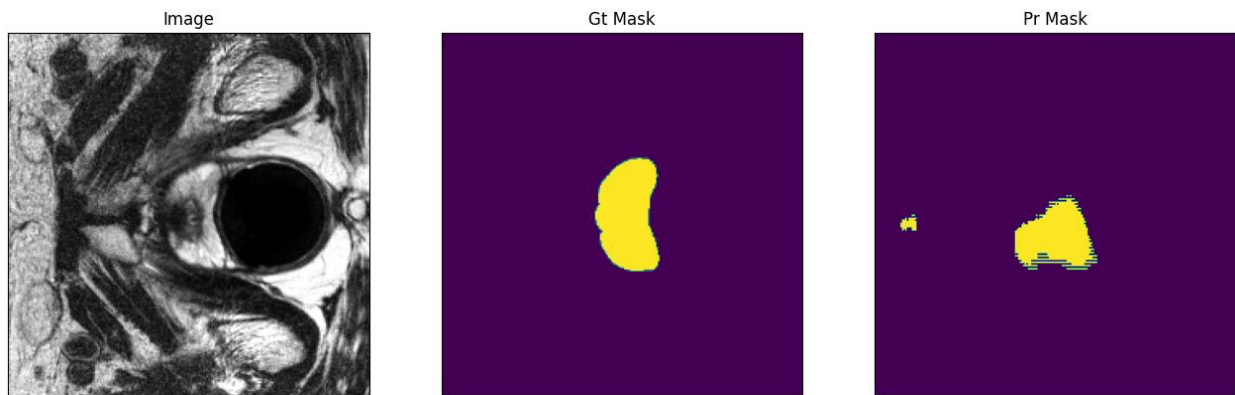


RESNET50: SITE B - 34 Epochs – LR 0.0001 – Loss: 0.15887 | mean iou_score: 0.7752 | mean f1-score: 0.85734

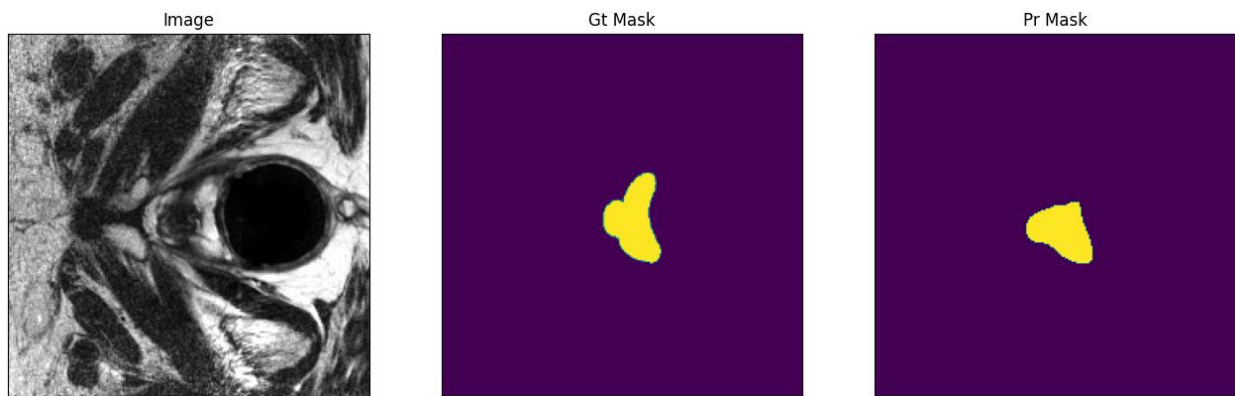
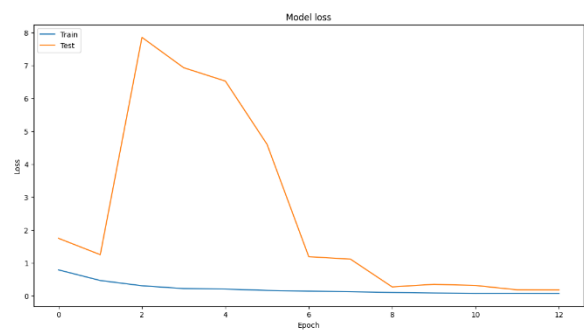
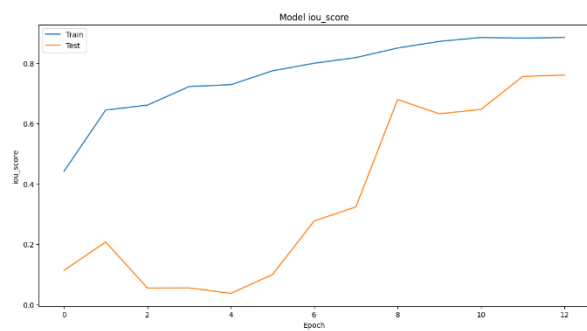


RESNET50: SITE B - 6 Epochs – LR 0.001 – Loss: 0.81414 | mean iou_score: 0.43038 | mean f1-score: 0.58981

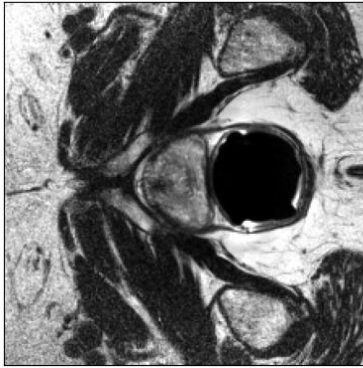




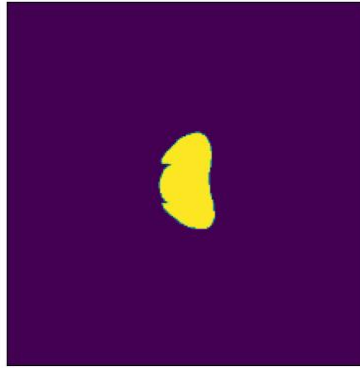
RESNET50: SITE B - 13 Epochs – LR 0.001 – Loss: 0.21704 | mean iou_score: 0.73127 | mean f1-score: 0.81857



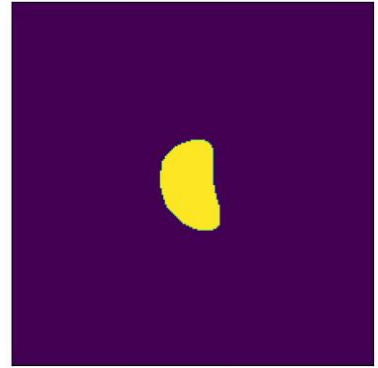
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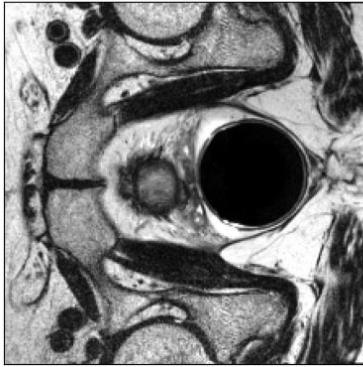
Gt Mask



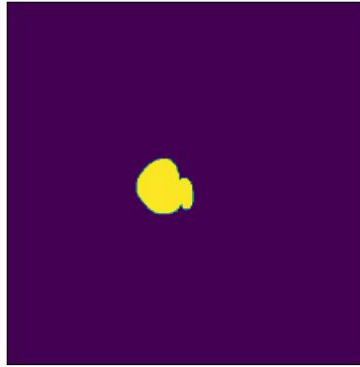
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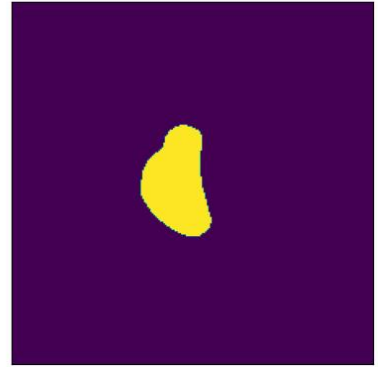
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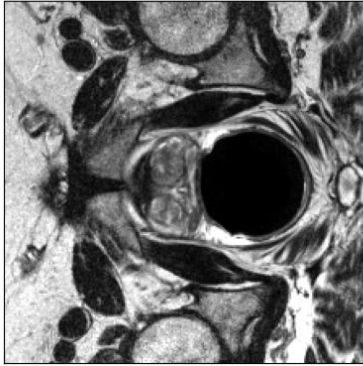
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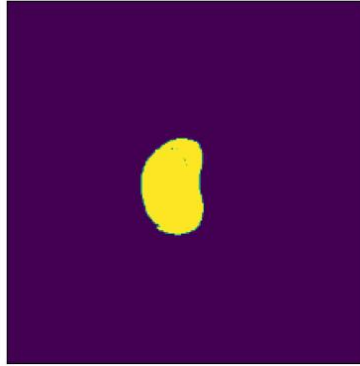
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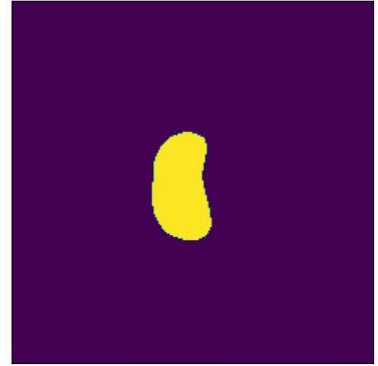
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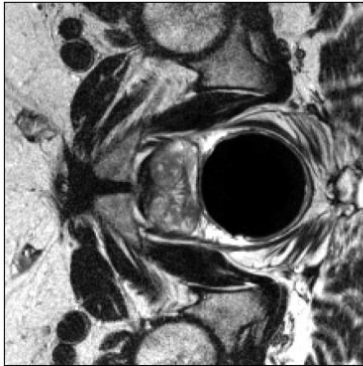
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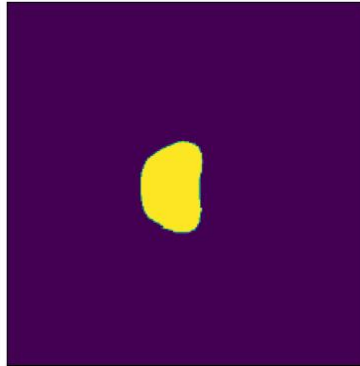
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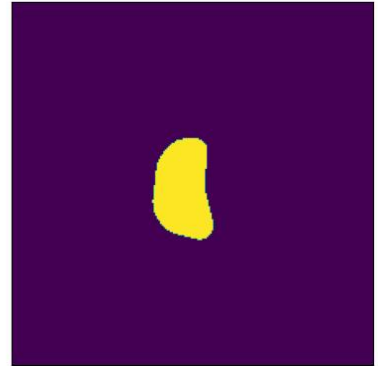
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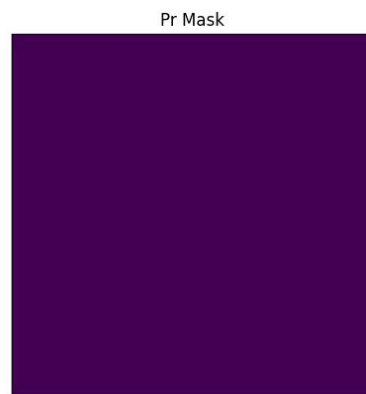
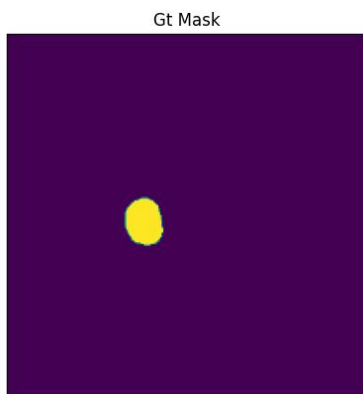
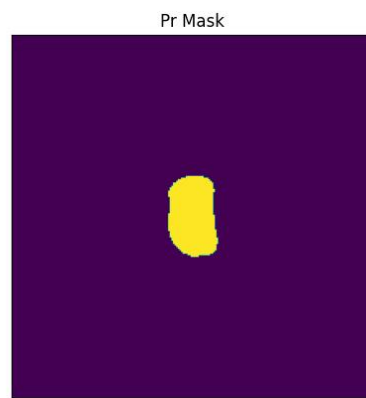
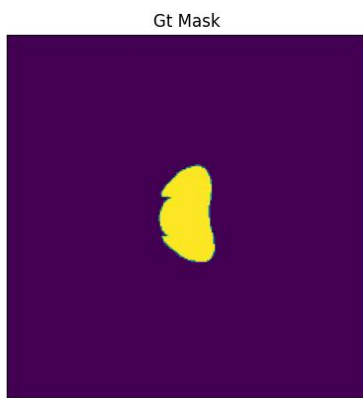
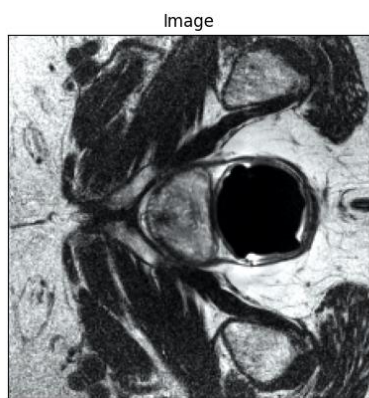
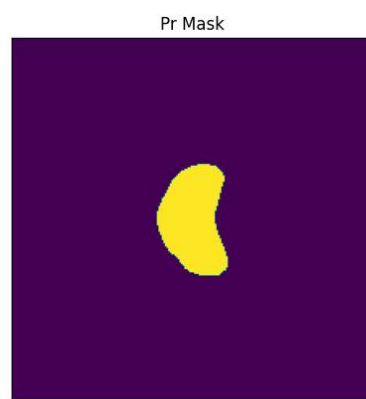
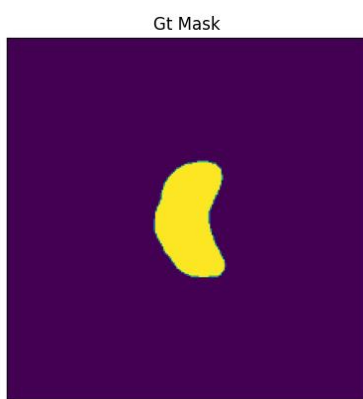
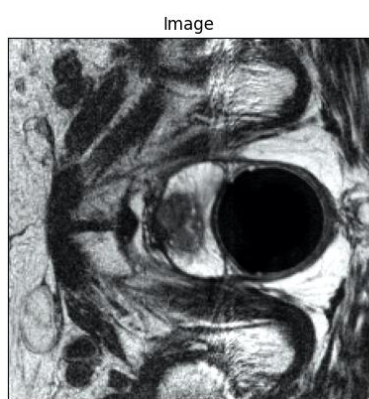
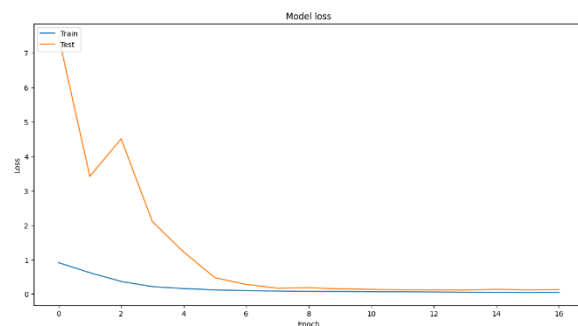
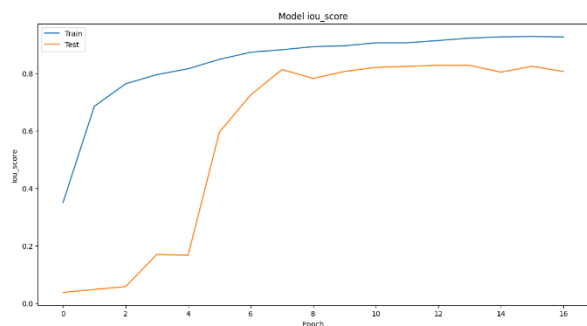


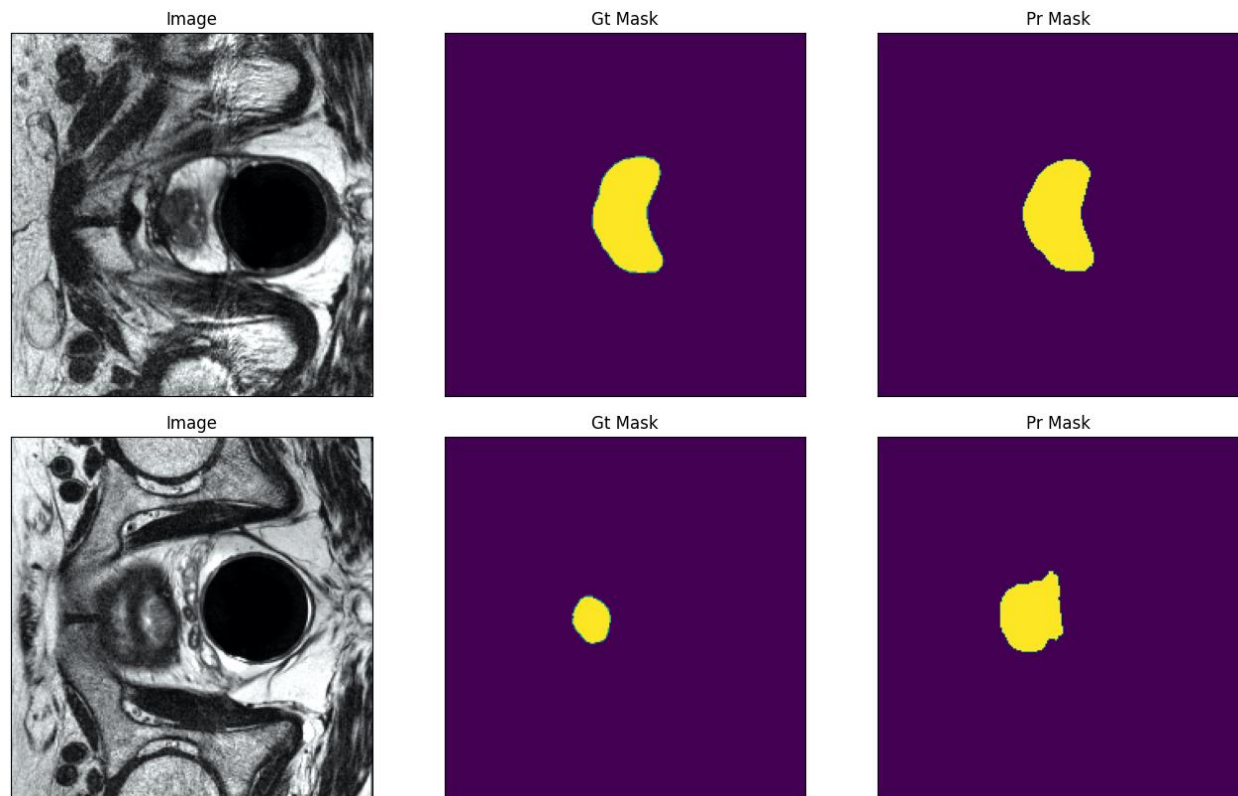
Pr Mask



EFFICIENTNETB0

EFFICIENTNETB0: SITE B – Epochs 17 – LR 0.001 – Loss: 0.16874 | mean iou_score: 0.77517 | mean f1-score: 0.85335





Comparison and Evaluation:

We conducted experiments to train and evaluate UNet models utilizing different backbone architectures on two distinct prostate MRI datasets from Site A and Site B. Our objective was to assess the effectiveness of these models in accurately segmenting prostate structures from MRI images. We evaluated their performance based on key metrics such as Intersection over Union (IoU), Mean F1 Score, and Loss.

Best Model Performance:

By analyzing the experimental results, we identified the best-performing model on each site. We considered factors such as epoch, learning rate, and performance metrics across the datasets to determine the optimal model for prostate MRI segmentation on Site A and Site B. The table below describes the result.

Site	Model	LR	Epochs	Loss	Mean iou_score	Mean f1-score
SITE A	VGG19	0.001	34	0.13674	0.79001	0.87815
	RESNET50	0.001	34	0.13569	0.79542	0.88085
	EFFICIENTNETB0	0.001	13	0.10565	0.84557	0.91272
SITE B	VGG19	0.0001	50	0.29092	0.62448	0.74278
	RESNET50	0.001	13	0.21704	0.73127	0.81857
	EFFICIENTNETB0	0.001	17	0.16874	0.77517	0.85335

For Site A:

- VGG19: Mean IoU Score = 0.79001, Mean F1-Score = 0.87815
- ResNet50: Mean IoU Score = 0.79542, Mean F1-Score = 0.88085
- EfficientNetB0: Mean IoU Score = 0.84557, Mean F1-Score = 0.91272

Here, the EfficientNetB0 model has the highest mean IoU score (0.84557) and mean F1-score (0.91272), indicating better performance compared to VGG19 and ResNet50.

For Site B:

- VGG19: Mean IoU Score = 0.62448, Mean F1-Score = 0.74278
- ResNet50: Mean IoU Score = 0.73127, Mean F1-Score = 0.81857
- EfficientNetB0: Mean IoU Score = 0.77517, Mean F1-Score = 0.85335

Here, the EfficientNetB0 model also has the highest mean IoU score (0.77517) and mean F1-score (0.85335), indicating better performance compared to VGG19 and ResNet50.

Therefore, the EfficientNetB0 model performed better on both Site A and Site B among the three models provided.

Cross-Site Test Datasets Evaluation:

The evaluation of cross-site testing aimed to assess the generalization capabilities of the models by testing their performance on datasets from sites different from their training data. In this case, we focused on evaluating the EfficientNetB0 model trained on Dataset Site B and tested it on Dataset Site A, and vice versa.

The performance metrics, particularly the mean IoU scores, demonstrate how well the model adapted to the different datasets. From the provided metrics, we observe that the EfficientNetB0 model trained on Dataset Site B achieved a mean IoU score of 0.70986 when tested on Dataset Site A, which is reasonably close to its performance on its native dataset. Similarly, when the model trained on Dataset Site A was tested on Dataset Site B, it achieved a mean IoU score of 0.47484. While this score is lower than its performance on its native dataset, it still demonstrates a reasonable level of generalization.

These results suggest that the EfficientNetB0 model exhibits decent generalization capabilities across different datasets, although there is some performance degradation when applied to datasets with different imaging conditions and characteristics. However, despite this degradation, the model still maintains a reasonable level of segmentation accuracy, as indicated by the mean IoU scores.

<i>Test Images\Best Model on Site mean iou_score</i>	Best Model Site A	Best Model Site B
<i>SITE A</i>	0.84557	0.70986
<i>SITE B</i>	0.47484	0.77517

Metrics for Site A using Model B

- Loss: 0.22654
- mean iou_score: 0.70986
- mean f1-score: 0.79992

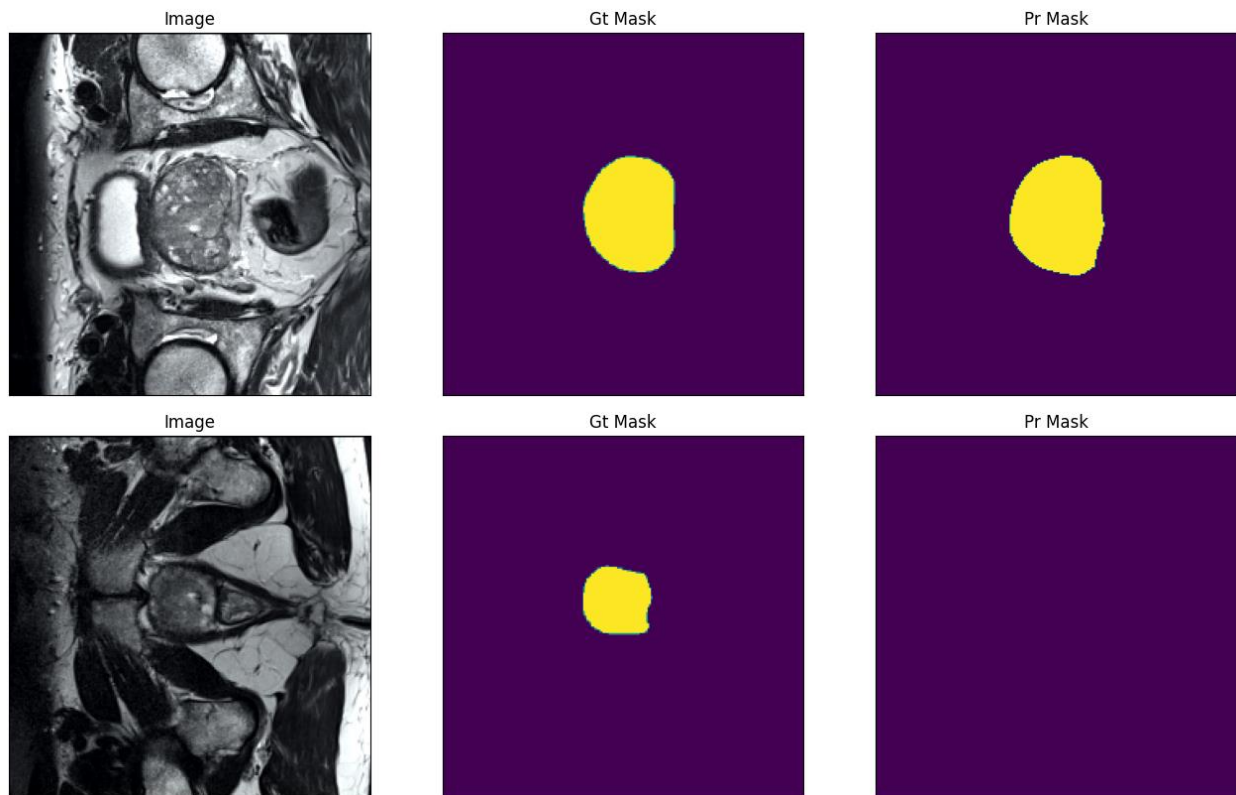
Metrics for Site B using Model A

- Loss: 0.44312
- mean iou_score: 0.47484
- mean f1-score: 0.59214

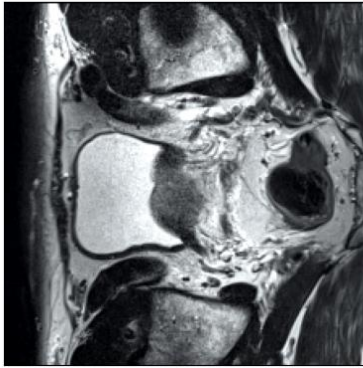
Visual Predictions

The visual predicted results from the test dataset further support these findings, showing that the EfficientNetB0 model trained on Dataset Site B can still provide meaningful segmentation results when applied to Dataset Site A, and vice versa. This suggests that the model can effectively learn and adapt to the variations in imaging conditions and structures present in different datasets, making it a robust choice for prostate MRI segmentation tasks across diverse datasets and sites.

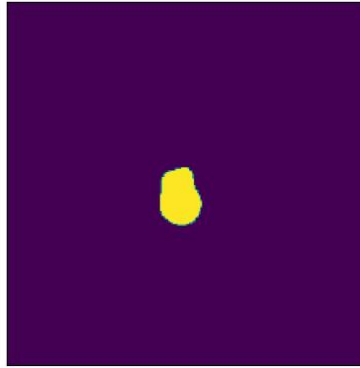
Below are the prediction results obtained from five random images taken from the test dataset of Site A using the EfficientNetB0 model trained on the dataset from Site B.



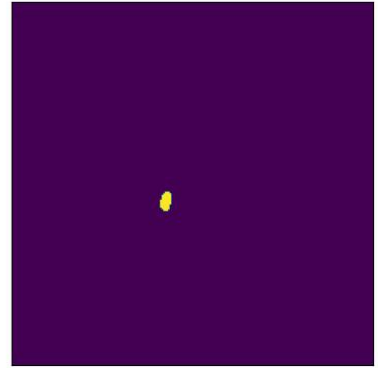
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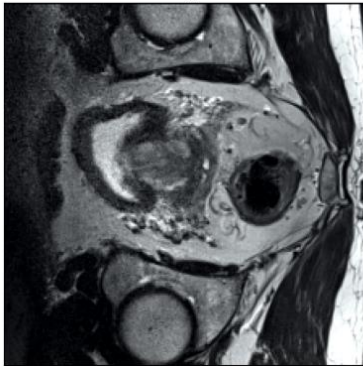
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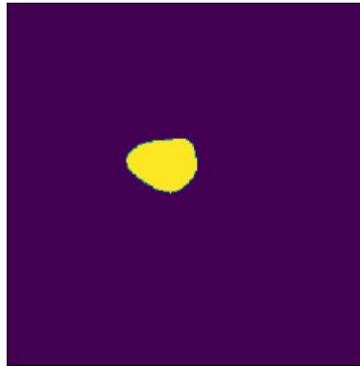
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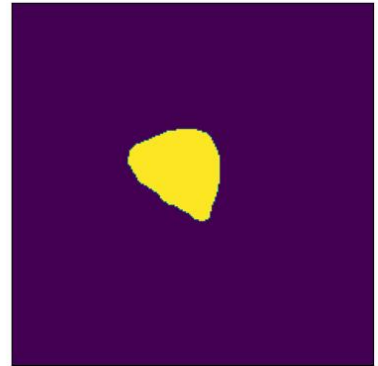
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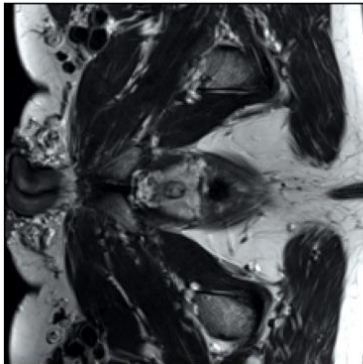
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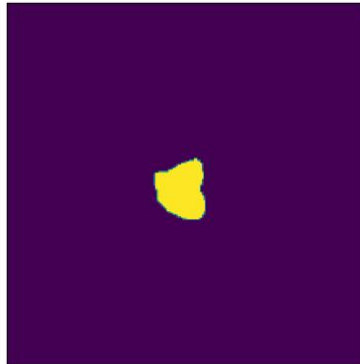
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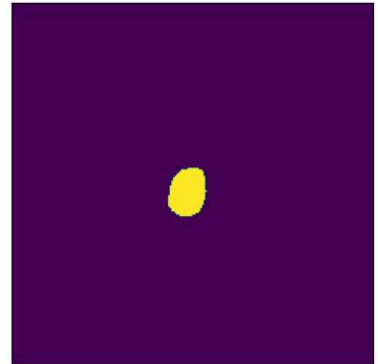
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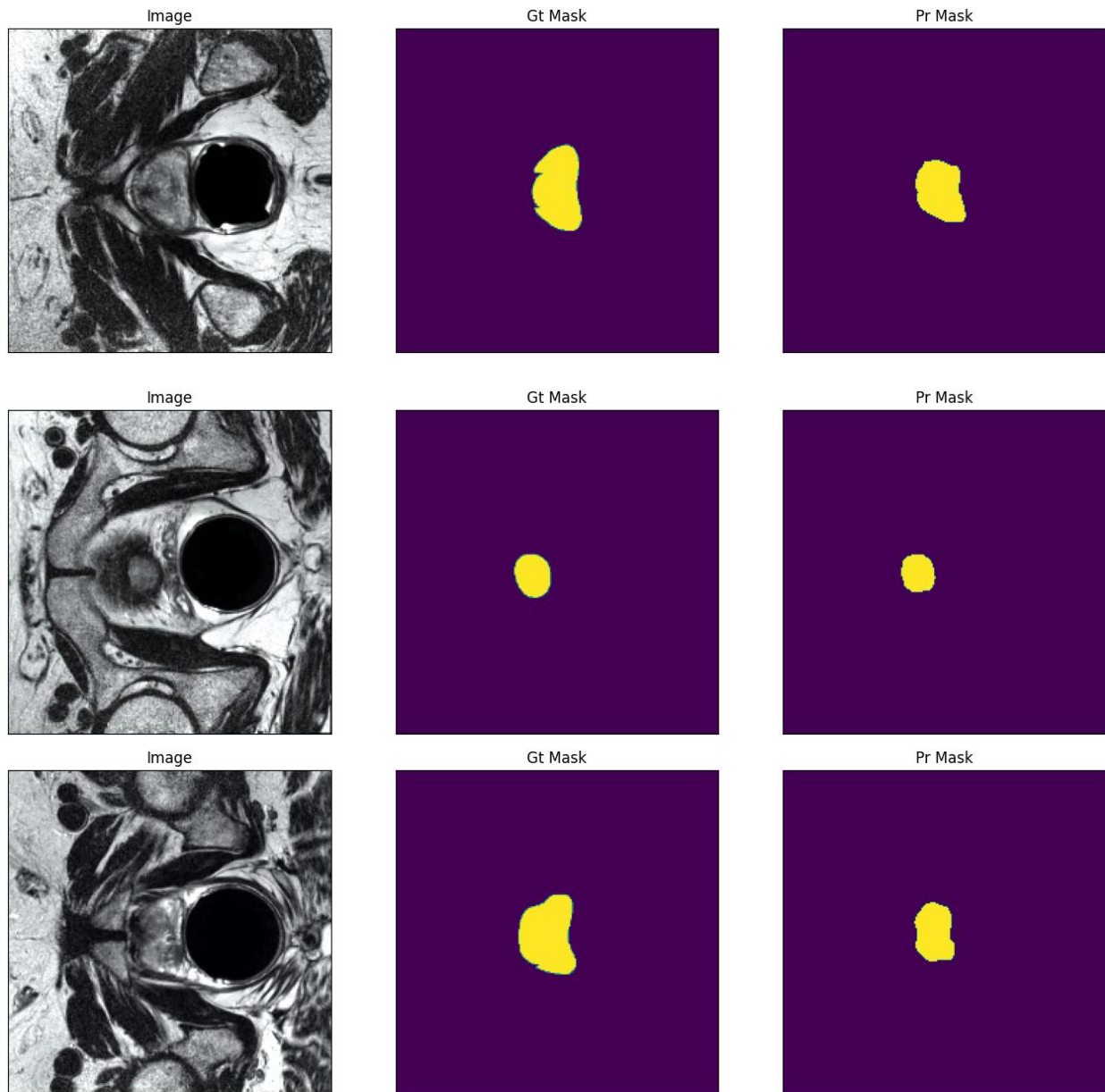
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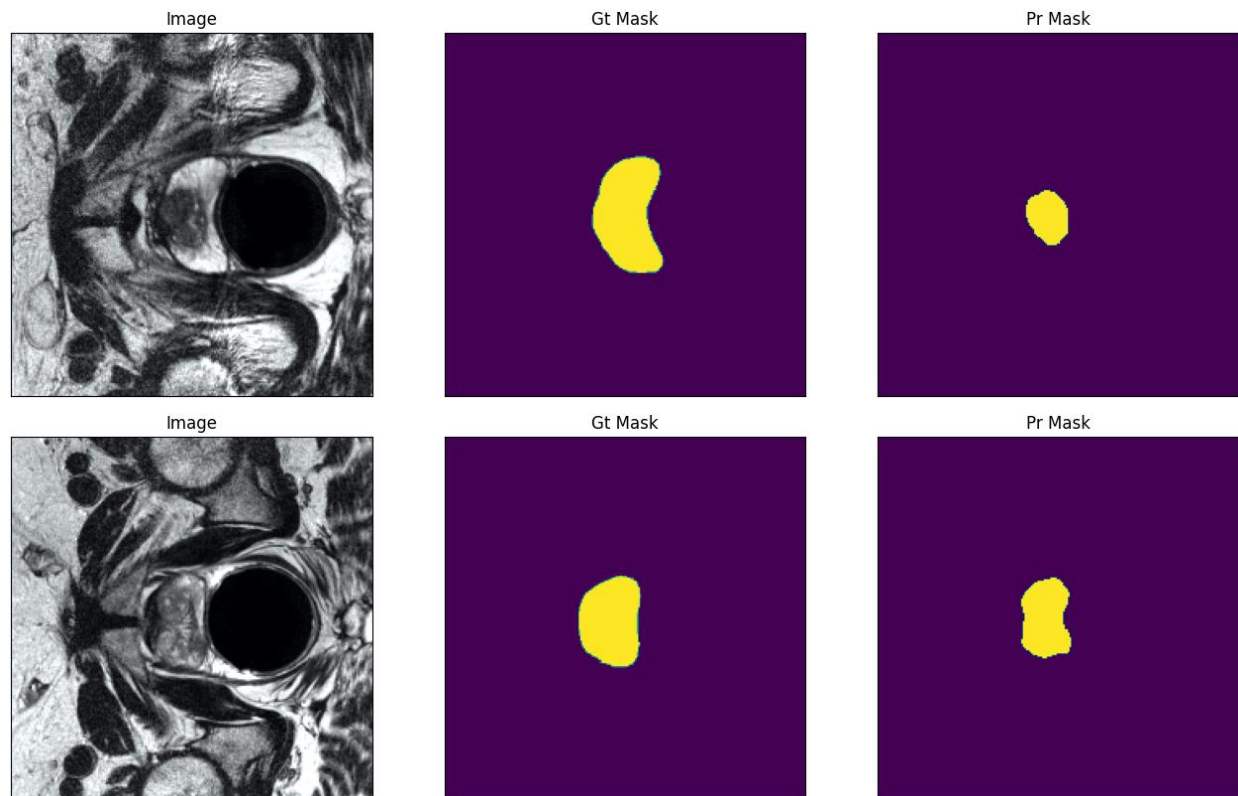


Pr Mask



Below are the prediction results obtained from five random images taken from the test dataset of Site B using the EfficientNetB0 model trained on the dataset from Site A.





Acknowledgement:

[https://github.com/qubvel/segmentation_models/blob/master/examples/binary%20segmentation%20\(camvid\).ipynb](https://github.com/qubvel/segmentation_models/blob/master/examples/binary%20segmentation%20(camvid).ipynb)

https://github.com/bnsreenu/python_for_microscopists/blob/master/231_234_BraTa2020_Unet_segmentation/232_brats2020_get_data_ready.py

Aldoj, N., Biavati, F., Michallek, F. *et al.* Automatic prostate and prostate zones segmentation of magnetic resonance images using DenseNet-like U-net. *Sci Rep* **10**, 14315 (2020).

<https://doi.org/10.1038/s41598-020-71080-0>